خدمتكم طريق خضناه لرضى الله

2021

ملخص

د.ياسين الحسن

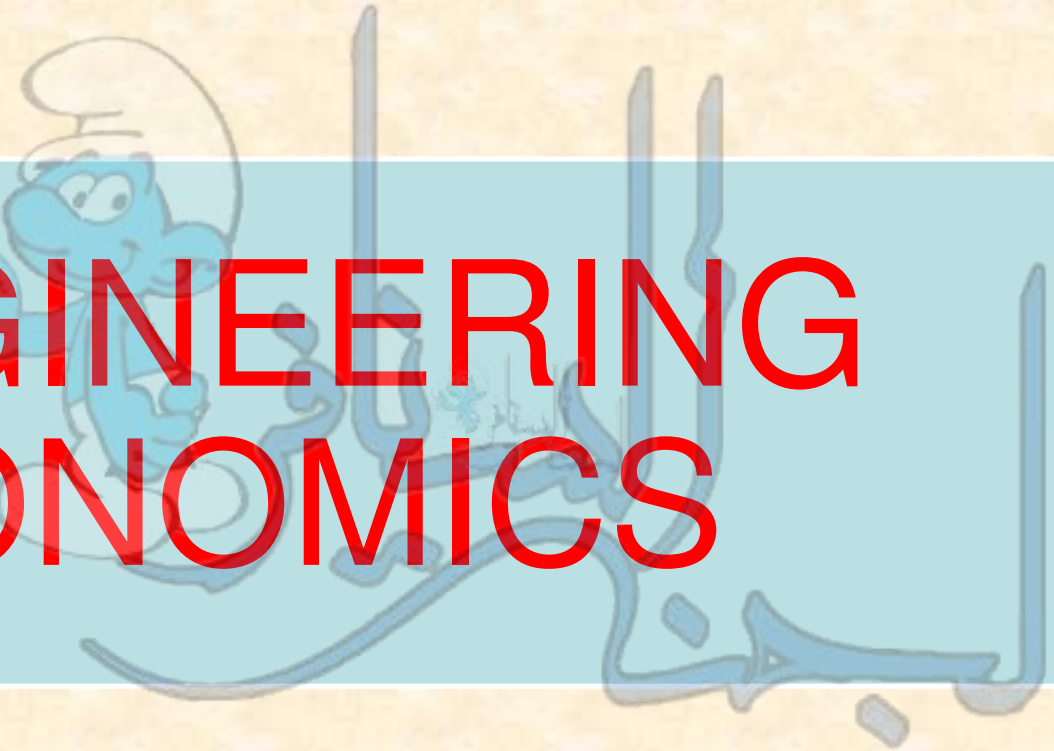
إقتصاد هندسي



0790640139

سنافر البوليتكنك

ENGINEERING ECONOMICS



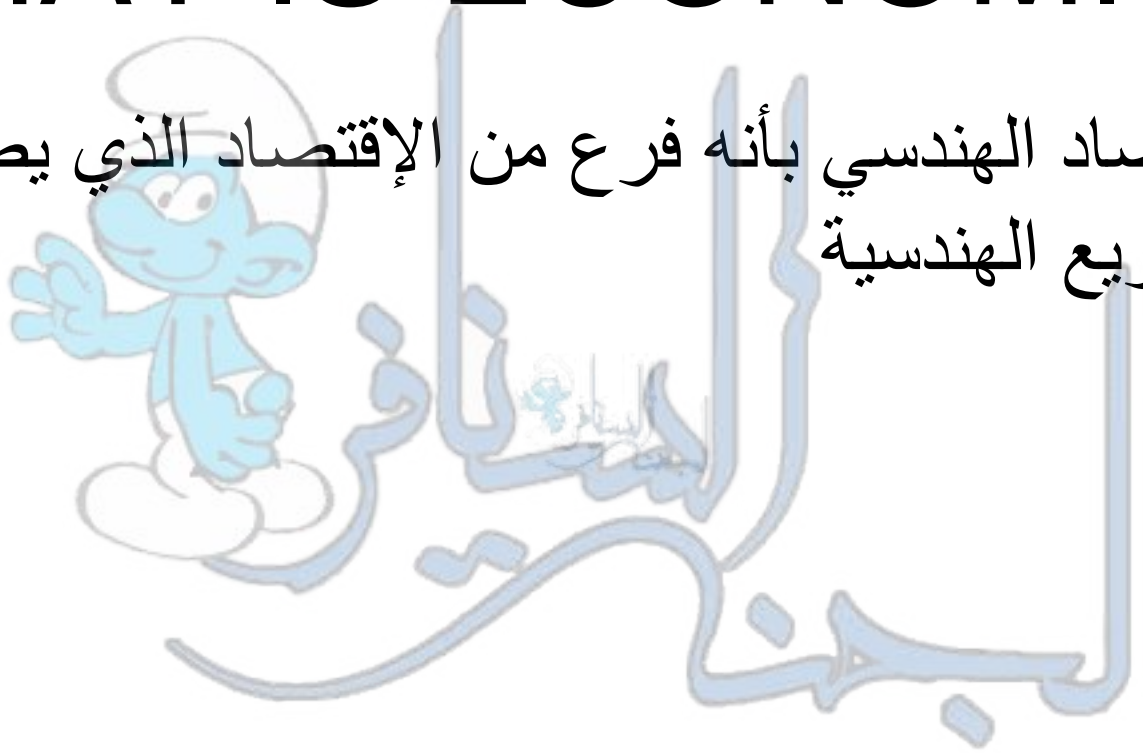
تم التحميل من شبكة المهندس المسلم
www.muslimengineer.info

WHAT IS ECONOMICS ?

The study of how limited resources is used to satisfy unlimited human wants

WHAT IS ECONOMICS ?

عرف الإقتصاد الهندسي بأنه فرع من الإقتصاد الذي يطبق
في المشاريع الهندسية

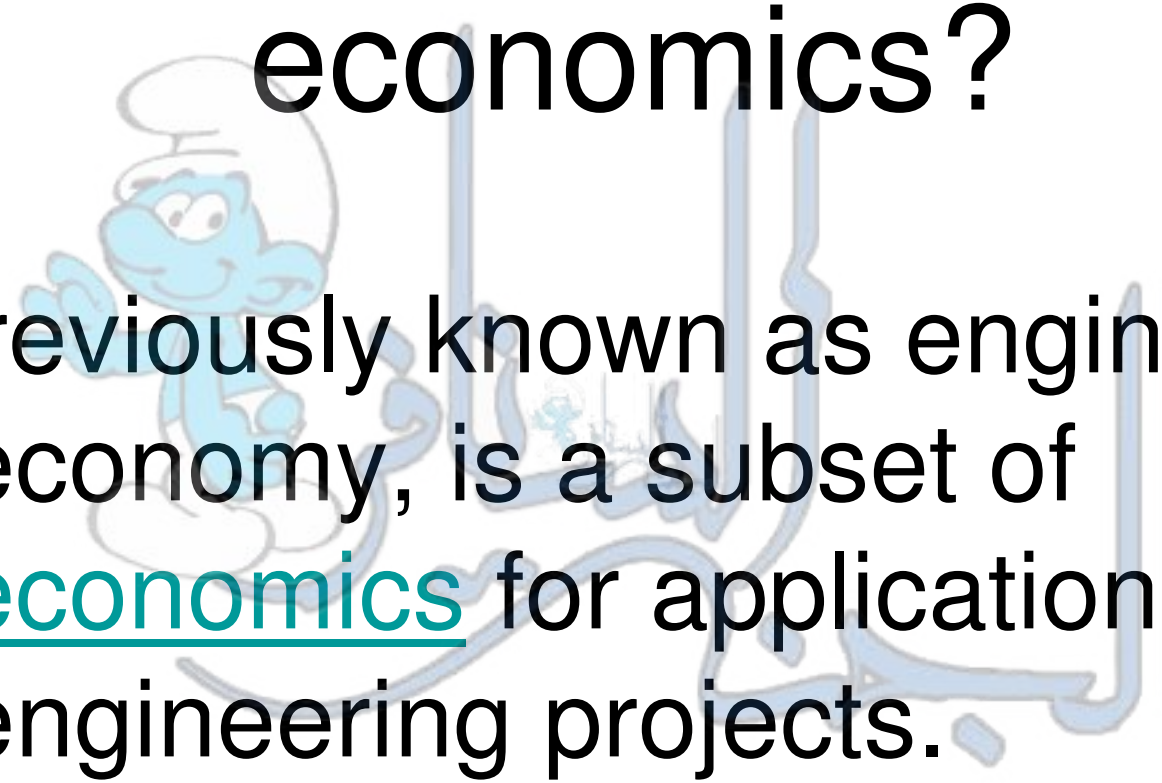


تم التحميل من شبكة المهندس المسلم
www.muslimengineer.info

WHAT IS Engineering economics?

,

previously known as engineering economy, is a subset of economics for application to engineering projects.



WHAT IS ECONOMICS ?

. Engineers seek solutions to problems, and the economic viability of each potential solution is normally considered along with the technical aspects

تم التحميل من شبكة المهندس المسلم
www.muslimengineer.info

Introduction to Engineering Economy

- التقييم المنهجي لتكاليف و منافع المشاريع التقنية المقترحة
- تستخدم مبادئ و مناهج الإقتصاد الهندسي في تحليل بدائل الإستخدامات البديلة للموارد المالية المتعلقة بالأصول الفيزيائية و الجوانب العملانية للمؤسسات الهندسية

Introduction to Engineering Economy

- The systematic evaluation of the costs and benefits of proposed technical projects
- The principles and methodology of engineering economy are utilized to analyze alternative uses of financial resources, particularly in relation to the physical assets and the operation of an organization.

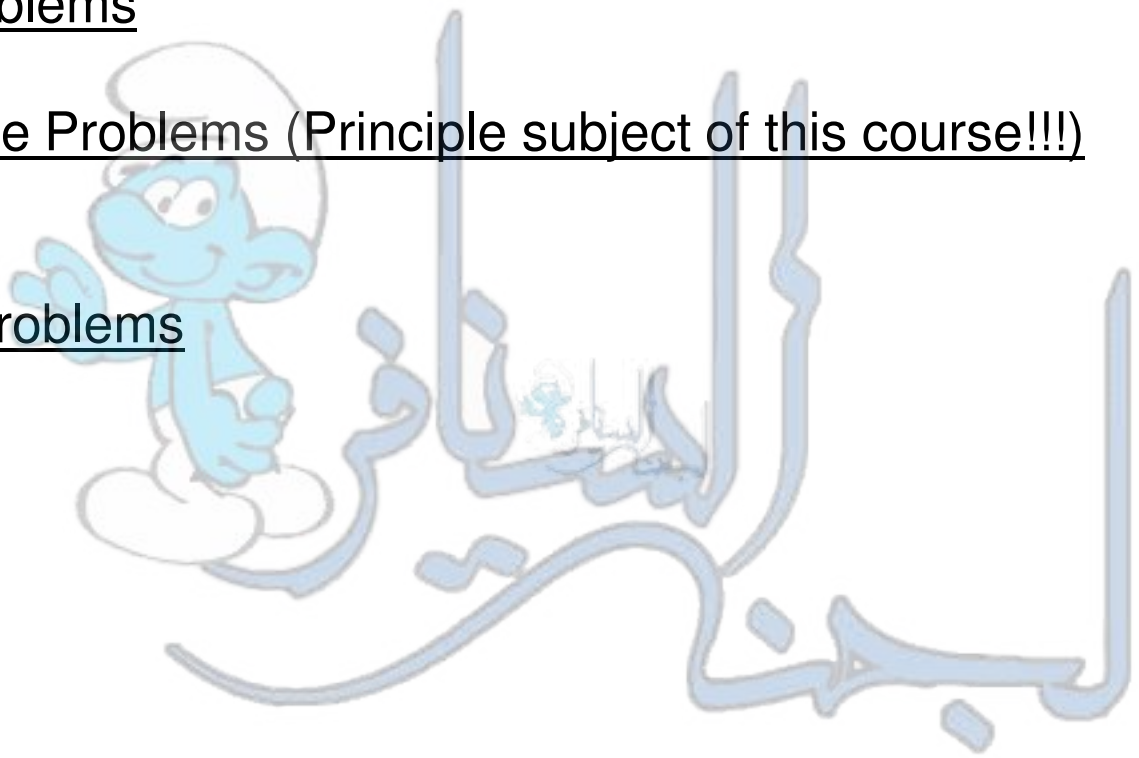
تم التحميل من شبكة المهندس المسلم
www.muslimengineer.info

Decision Making and Problem Solving

Simple Problems

Intermediate Problems (Principle subject of this course!!!)

Complex Problems



Decision Making and Problem Solving

Simple Problems

- Can be analyzed in one's head without extensive analysis
  - Do I buy a semester parking pass or use parking meters?
  - Should I fix my car?

Intermediate Problems (Principle subject of this course!!!)

- They are sufficiently important to justify serious thought and action
- They can't be worked in one's head; must be organized
- The economic aspects are significant component in the analysis leading to a decision

Complex Problems

- Represent a mixture of economic, political, and humanistic elements
  - Selection of a president in USA
  - Building a nuclear power plant

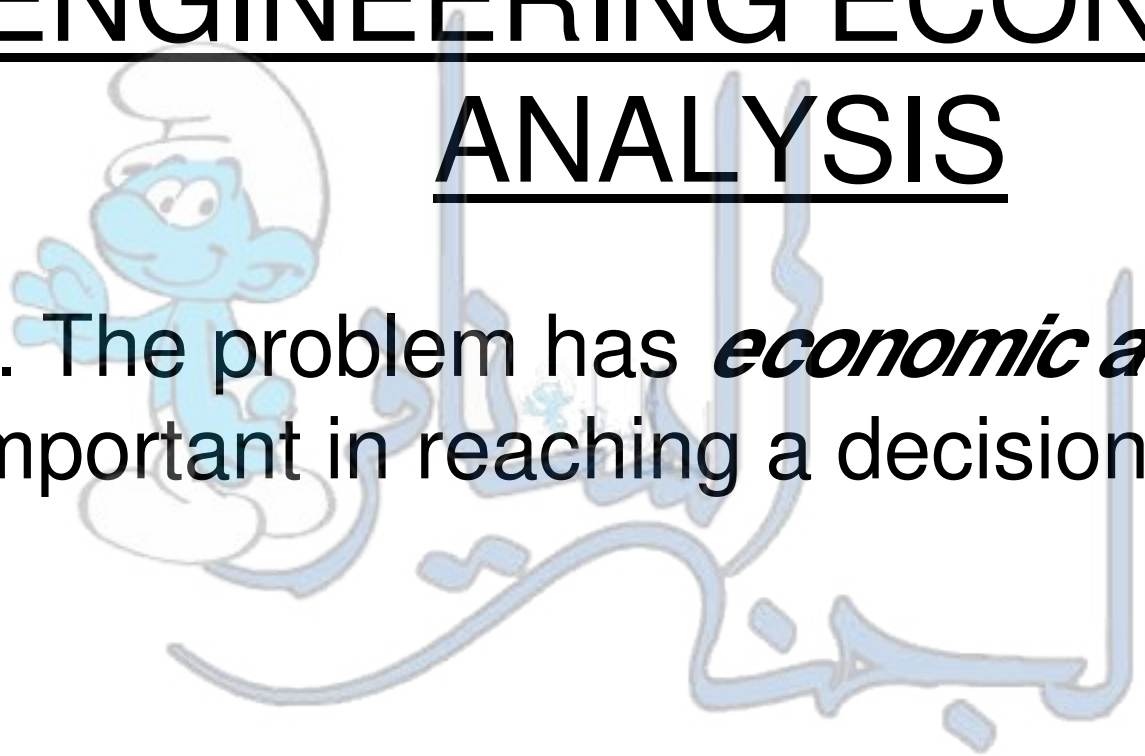
تم التحميل من شبكة المهندس المسلم
www.muslimengineer.info

THE ROLE OF ENGINEERING ECONOMIC ANALYSIS

- 1. The problem is ***important enough*** to justify our giving it serious thought and effort.
- 2. The problem can't be worked in one's head-that is, a careful analysis ***requires that we organize*** the problem and all the various consequences, and this is just too much to be done all at once.

THE ROLE OF ENGINEERING ECONOMIC ANALYSIS

- 3. The problem has ***economic aspects*** important in reaching a decision.

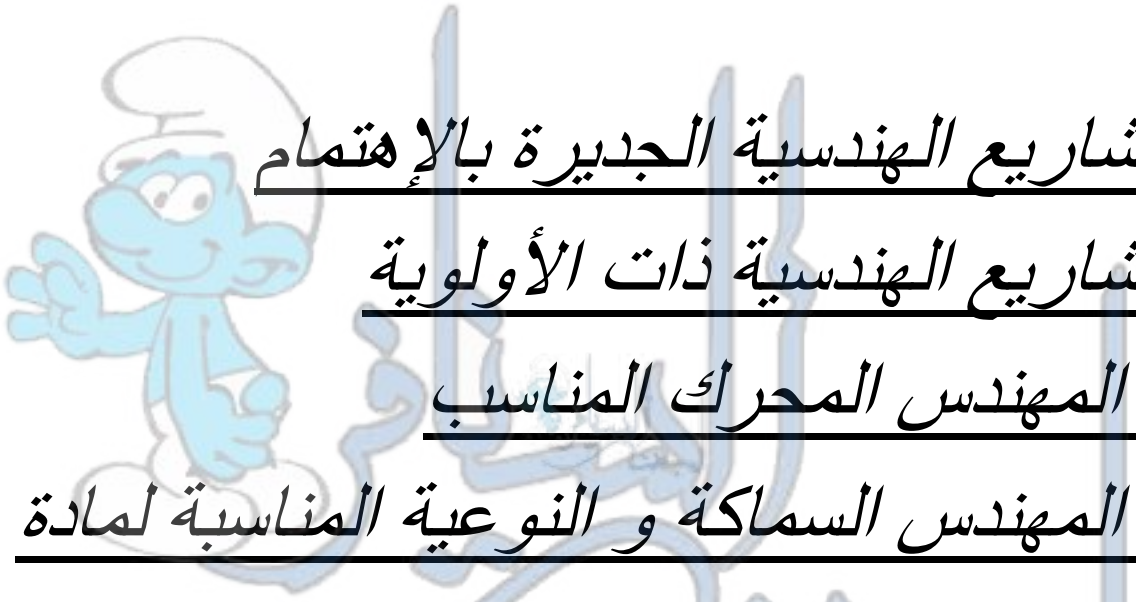


تم التحميل من شبكة المهندس المسلم
www.muslimengineer.info

Examples of Engineering Economic Analysis

- Engineering economic analysis focuses on costs, revenues, and benefits that occur at different times. For example, when a civil engineer "designs a road, a dam, or a building, the construction costs occur in the near future; the benefits to users begin only when construction is finished, but then the benefits continue for a long time.

Examples of Engineering Economic Analysis



- ما هي المشاريع الهندسية الجديرة بالإهتمام
- ما هي المشاريع الهندسية ذات الأولوية
- هل اختار المهندس المحرك المناسب
- هل اختار المهندس السماكة و النوعية المناسبة لمادة العزل
- هل وازن بين بديلين احدهما مكلف و آمن و الآخر اقل كلفة و اقل أمنا

تم التحميل من شبكة المهندس المسلم
www.muslimengineer.info

Rational Decision Making

1. Recognize the problem
   - "I need a place to live this term."
2. Define the Goal or Objective
   - "I'll find a nice apartment that is not too expensive."
3. Assemble Relevant Data
   - "I need information on rent, utilities, apartment age, parking, driving time to UF, driving time to shopping, the neighborhood, other amenities provided (swimming, table tennis, etc.)."
4. Identify Feasible Alternatives
   - "I'll use the Yellow Pages, information from friends, apartment finding services, information from UF, the local newspaper, and my personal experience, to look for apartments."

Rational Decision Making

5. Select Criterion to Determine the Best Alternative
   - "Most important is rent plus utilities cost. I'm also very concerned about driving time to UF, and the kind of neighborhood the apartment is in."

6. Construct the model
   - "I'll make a spreadsheet. The rows will be the apartment choices, the columns the evaluation criteria. Then I'll try to fill in the interactions between the apartments and the criteria."
   - This includes determining cash flows for engineering economic analysis!!!

7. Predict Outcomes of Each Alternative
   - "I'll fill in the estimated costs for the spreadsheet and rate the amenities, driving time, etc."

8. Choose the Best Alternative
   - "Apartment C looks cheapest, but I don`t like the neighborhood. If I pay $50 more per month for Apartment B I get a nicer neighborhood, and a 15-minute drive to UF. Maybe I'll choose Apartment B."

تم التحميل من شبكة المهندس المسلم
www.muslimengineer.info

Rational Decision Making

9. Audit the Results

- "Did I make a good choice"
- "After living in Apartment B for six months, I am very happy with my choice!"
- But this certainly isn't the case every time!!

Rational Decision Making

١. التعرف على المشكلة
٢. تحديد الهدف
٣. تجميع البيانات ذات الصلة
٤. تحديد البدائل الممكنة
٥. اختيار المعايير المناسبة لتحديد افضل البدائل
٦. بناء النموذج
٧. توقع النتائج المتعلقة بكل من البدائل
٨. اختيار البديل الأفضل
٩. مراجعة النتائج

تم التحميل من شبكة المهندس المسلم
www.muslimengineer.info

PRINCIPLES OF ENGINEERING ECONOMY

1. Develop the Alternatives;
2. Focus on the Differences;
3. Use a Consistent Viewpoint;
4. Use a Common Unit of Measure;
5. Consider All Relevant Criteria;
6. Make Uncertainty Explicit;
7. Revisit Your Decisions

DEVELOP THE ALTERNATIVES

The final choice (decision) is among alternatives. The alternatives need to be identified and then defined for subsequent analysis.



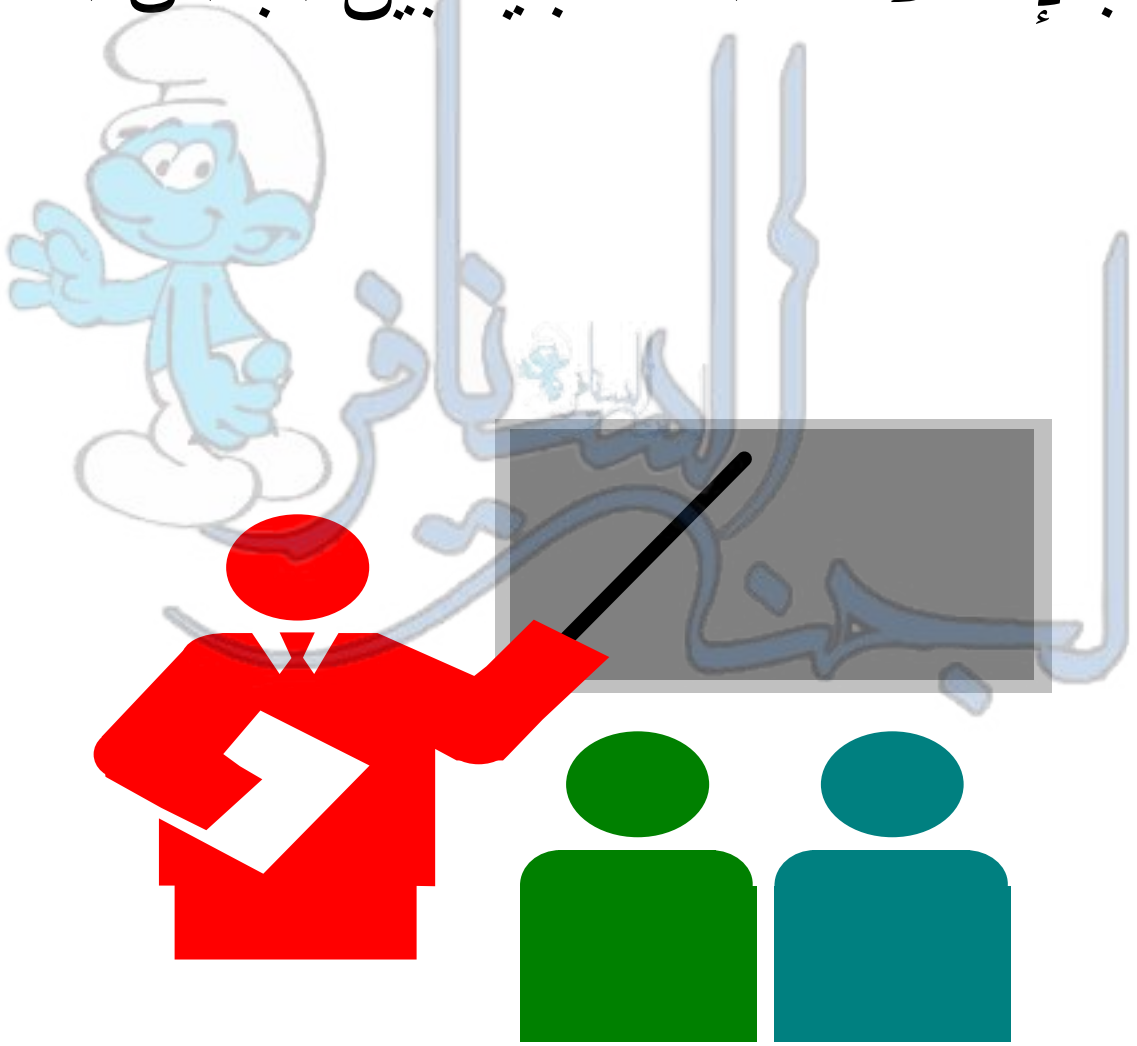
تم التحميل من شبكة المهندس المسلم
www.muslimengineer.info

Examples of Engineering Economic Analysis

- Engineering economic analysis is used to answer many different questions. - ***Which engineering projects are worthwhile?*** Has the mining or petroleum engineer shown that the mineral or oil deposit is worth developing?
- ***Which engineering projects should have a higher priority?*** Has the industrial engineer shown which factory improvement projects should be funded

FOCUS ON THE DIFFERENCES

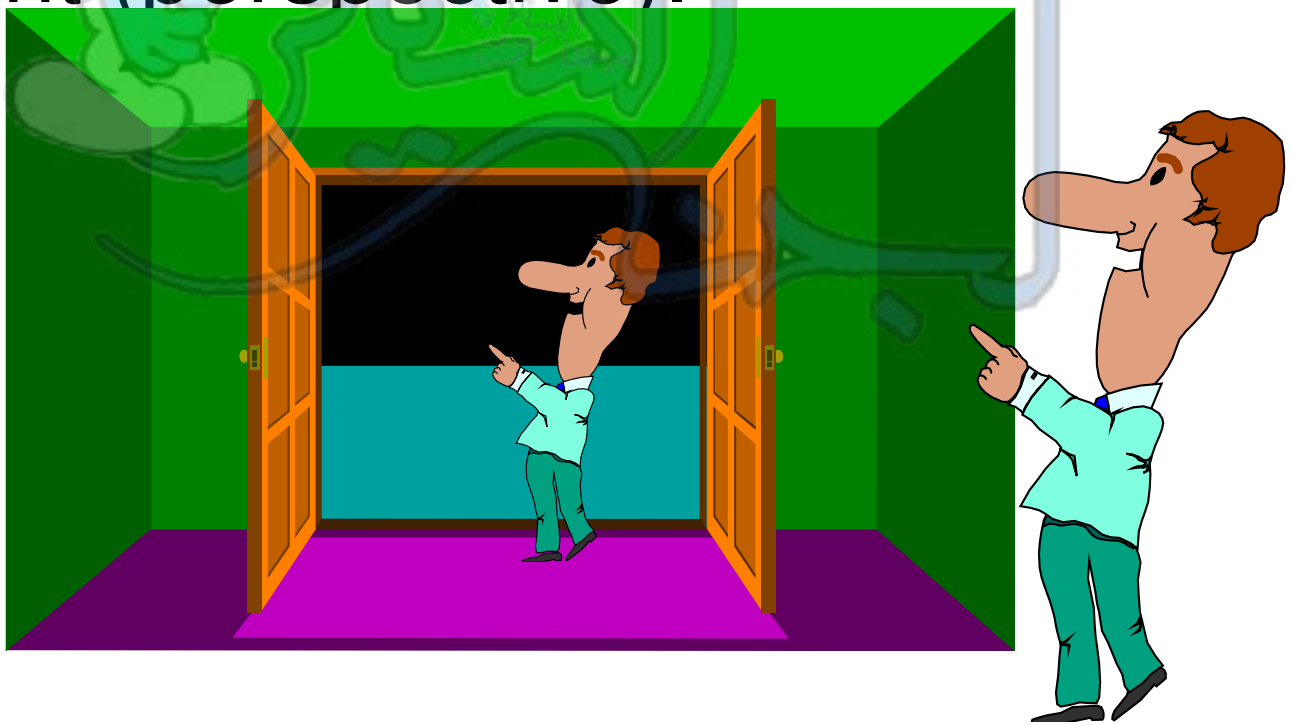
يجب الإهتمام بالإختلافات المستقبلية بين البدائل المقارنة



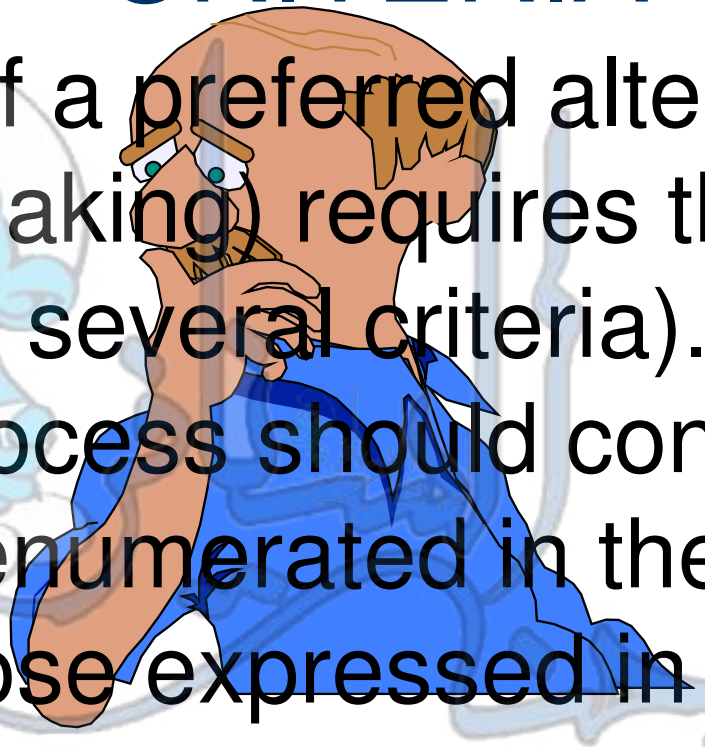
تم التحميل من شبكة المهندس المسلم
www.muslimengineer.info

USE A CONSISTENT VIEWPOINT

The prospective outcomes of the alternatives, economic and other, should be consistently developed from a defined viewpoint (perspective).

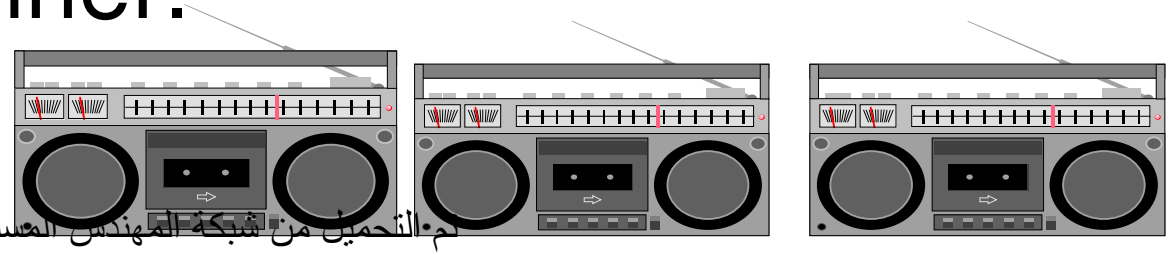


CONSIDER ALL RELEVANT CRITERIA



Selection of a preferred alternative (decision making) requires the use of a criterion (or several criteria). The decision process should consider the outcomes enumerated in the monetary unit and those expressed in some other unit of measurement or made explicit in a descriptive manner.

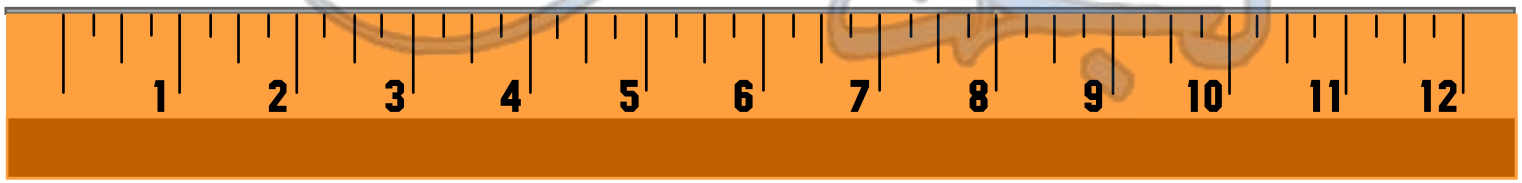


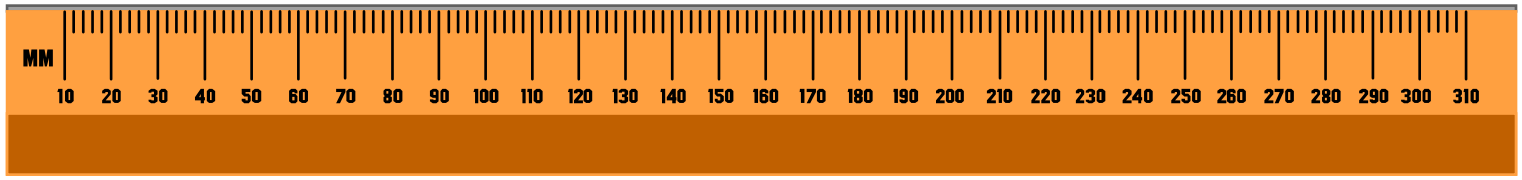


تم التحميل من شبكة المهندس المسلم
www.muslimengineer.info

USE A COMMON UNIT OF MEASURE

Using a common unit of measurement to enumerate as many of the prospective outcomes as possible will make easier the analysis and comparison of alternatives.





MAKE UNCERTAINTY EXPLICIT

Uncertainty is inherent in projecting (or estimating) the future outcomes of the alternatives and should be recognized in their analysis and comparison.



تم التحميل من شبكة المهندس المسلم
www.muslimengineer.info

MAKE UNCERTAINTY EXPLICIT



- ماذا يحدث اذا زادت التكاليف
- ماذا يحدث اذا انخفضت التكاليف
- ماذا يحدث اذا انخفضت الإيرادات
- ماذا يحدث اذا انخفضت الإيرادات و زادت التكاليف

REVISIT YOUR DECISIONS

Improved decision making results from an adaptive process; to the extent practicable, the initial projected outcomes of the selected alternative should be subsequently compared with actual results achieved.



تم التحميل من شبكة المهندس المسلم
www.muslimengineer.info

Time Value of Money

- The following are reasons why $1000 today is
- “worth” more than $1000 one year from today:
- 1. Inflation
- 2. Risk
- 3. Cost of money

INTEREST

The fee that a borrower pays to a lender for the use of his or her money.

INTEREST RATE

The percentage of money being borrowed that is paid to the lender on some time basis.

تم التحميل من شبكة المهندس المسلم
www.muslimengineer.info

Concept of Interest

If you won the lotto, would you rather get $1 Million now or $50,000 for 25 years?

What about automobile and home financing? What type of financing makes more economic sense?

Interest: Money paid for the use of borrowed money.

- Put simply, interest is the rental charge for **using** an asset over some period of time and then, returning the asset in the same conditions as we received it.

  → In project financing, the asset is usually ***money***

Why Interest exist?

Taking the lender's view of point:

- **Risk**: Possibility that the borrower will be unable to pay
- **Inflation**: Money repaid in the future will "value" less
- **Transaction Cost**: Expenses incurred in preparing the loan agreement
- **Opportunity Cost**: Committing limited funds, a lender will be unable to take advantage of other opportunities.
- **Postponement of Use**: Lending money, postpones the ability of the lender to use or purchase goods.

From the borrowers perspective ….

Interest represents a cost !

تم التحميل من شبكة المهندس المسلم
www.muslimengineer.info

Simple Interest

Simple Interest is also known as the Nominal Rate of Interest

- Annualized percentage of the amount borrowed (principal) which is paid for the use of the money for some period of time.

Suppose you invested $1,000 for one year at 6% simple rate; at the end of one year the investment would yield:

$$\$1,000 + \$1,000(0.06) = \$1,060$$

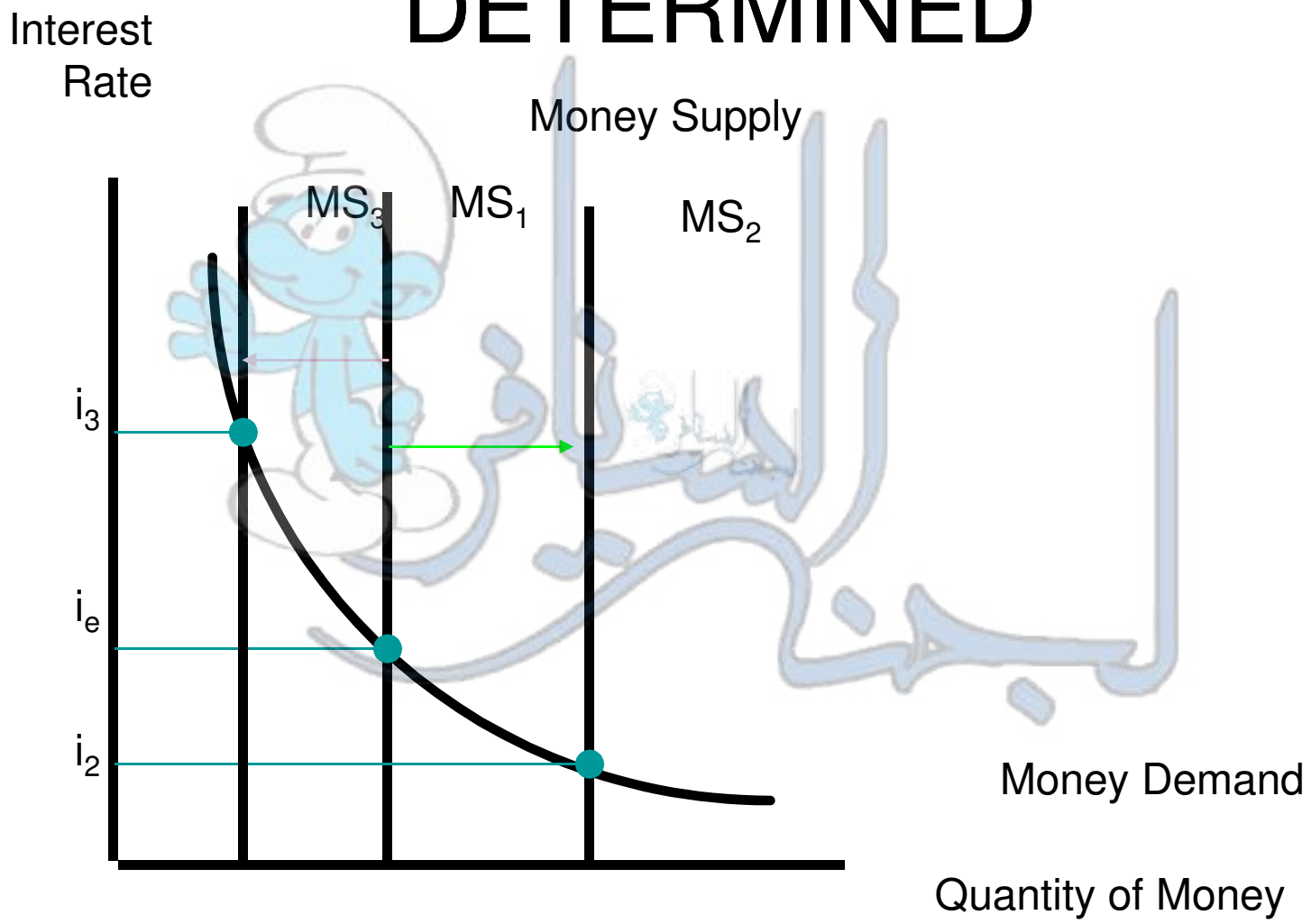
This means that each year interest gives $60

How much will you earn (including principal) after 3 years?

$$\$1,000 + \$1,000(0.06) + \$1,000(0.06) + \$1,000(0.06) = \$1,180$$

Note that for each year, the interest earned is only calculated over $1,000. Does this mean that you could draw the $60 earned at the end of each year?

HOW INTEREST RATE IS DETERMINED



تم التحميل من شبكة المهندس المسلم
www.muslimengineer.info

SIMPLE INTEREST

- The total interest earned or charged is linearly proportional to the initial amount of the loan (principal), the interest rate and the number of interest periods for which the principal is committed.
- When applied, total interest "I" may be found by

$$I = (P) (N) (i), \text{ where}$$

  - P = principal amount lent or borrowed
  - N = number of interest periods (e.g., years)
  - i = interest rate per interest period

Terms

In most situations, the percentage is not paid at the end of the period, where the interest earned is instead added to the original amount (principal). In this case, interest earned from previous periods is part of the basis for calculating the new interest payment.

This "adding up" defines the concept of **Compounded Interest**

Now assume you invested $1,000 for two years at 6% compounded annually;

At the end of one year the investment would yield:

$$\$1{,}000 + \$1{,}000 (0.06) = \$1{,}060 \quad \text{or} \quad \$1{,}000 (1 + 0.06)$$

Since interest is compounded annually, at the end of the second year the investment would be worth:

$$[\underbrace{\$1{,}000 (1 + 0.06)}_{\text{Principal and Interest for First Year}}] + [\underbrace{\$1{,}000 (1 + 0.06) (0.06)}_{\text{Interest for Second Year}}] = \$1{,}124$$

Factorizing:

$$\$1{,}000 (1 + 0.06) (1 + 0.06) = \$1{,}000 (1 + 0.06)^2 = \$1{,}124$$

How much this investment would yield at the end of year 3?

تم التحميل من شبكة المهندس المسلم
www.muslimengineer.info

ECONOMIC EQUIVALENCE

- Established when we are indifferent between a future payment, or a series of future payments, and a present sum of money .
- Considers the comparison of alternative options, or proposals, by reducing them to an equivalent basis, depending on:
  - interest rate;
  - amounts of money involved;
  - timing of the affected monetary receipts and/or expenditures;
  - manner in which the interest , or profit on invested capital is paid and the initial capital is recovered.

ECONOMIC EQUIVALENCE FOR FOUR REPAYMENT PLANS OF AN $8,000 LOAN

- Plan #1: $2,000 of loan principal plus 10% of BOY principal paid at the end of year; interest paid at the end of each year is reduced by $200 (i.e., 10% of remaining principal)

Year	Amount Owed at beginning of Year (BOY)	Interest Accrued for Year	Total Money owed at end of Year	Principal Payment	Total end of Year Payment
1	$8,000	$800	$8,800	$2,000	$2,800
2	$6,000	$600	$6,600	$2,000	$2,600
3	$4,000	$400	$4,400	$2,000	$2,400
4	$2,000	$200	$2,200	$2,000	$2,200

Total interest paid ($2,000) is 10% of total dollar-years ($20,000)

تم التحميل من شبكة المهندس المسلم
www.muslimengineer.info

- Plan #2: $0 of loan principal paid until end of fourth year; $800 interest paid at the end of each year

Year	Amount Owed at beginning of Year (BOY)	Interest Accrued for Year	Total Money owed at end of Year	Principal Payment	Total end of Year Payment
1	$8,000	$800	$8,800	$0	$800
2	$8,000	$800	$8,800	$0	$800
3	$8,000	$800	$8,800	$0	$800
4	$8,000	$800	$8,800	$8,000	$8,800

Total interest paid ($3,200) is 10% of total dollar-years ($32,000)

- Plan #3: $2,524 paid at the end of each year; interest paid at the end of each year is 10% of amount owed at the beginning of the year.

Year	Amount Owed at beginning of Year (BOY)	Interest Accrued for Year	Total Money owed at end of Year	Principal Payment	Total end of Year Payment
1	$8,000	$800	$8,800	$1,724	$2,524
2	$6,276	$628	$6,904	$1,896	$2,524
3	$4,380	$438	$4,818	$2,086	$2,524
4	$2,294	$230	$2,524	$2,294	$2,524

Total interest paid ($2,096) is 10% of total dollar-years ($20,950)

تم التحميل من شبكة المهندس المسلم
www.muslimengineer.info

- Plan #4: No interest and no principal paid for first three years. At the end of the fourth year, the original principal plus accumulated (compounded) interest is paid.

Year	Amount Owed at beginning of Year (BOY)	Interest Accrued for Year	Total Money owed at end of Year	Principal Payment	Total end of Year Payment
1	$8,000	$800	$8,800	$0	$0
2	$8,800	$880	$9,680	$0	$0
3	$9,680	$968	$10,648	$0	$0
4	$10,648	$1,065	$11,713	$8,000	$11,713

Total interest paid ($3,713) is 10% of total dollar-years ($37,128)

CASH FLOW DIAGRAMS / TABLE NOTATION

i = effective interest rate per interest period

N = number of compounding periods (e.g., years)

P = present sum of money; the equivalent value of one or more cash flows at the present time reference point

F = future sum of money; the equivalent value of one or more cash flows at a future time reference point

A = end-of-period cash flows (or equivalent end-of-period values) in a uniform series continuing for a specified number of periods, starting at the end of the first period and continuing through the last period

G = uniform gradient amounts -- used if cash flows increase by a constant amount in each period

CASH FLOW DIAGRAM NOTATION



1

Time scale with progression of time moving from left to right; the numbers represent time periods (e.g., years, months, quarters, etc...) and may be presented within a time interval or at the end of a time interval.

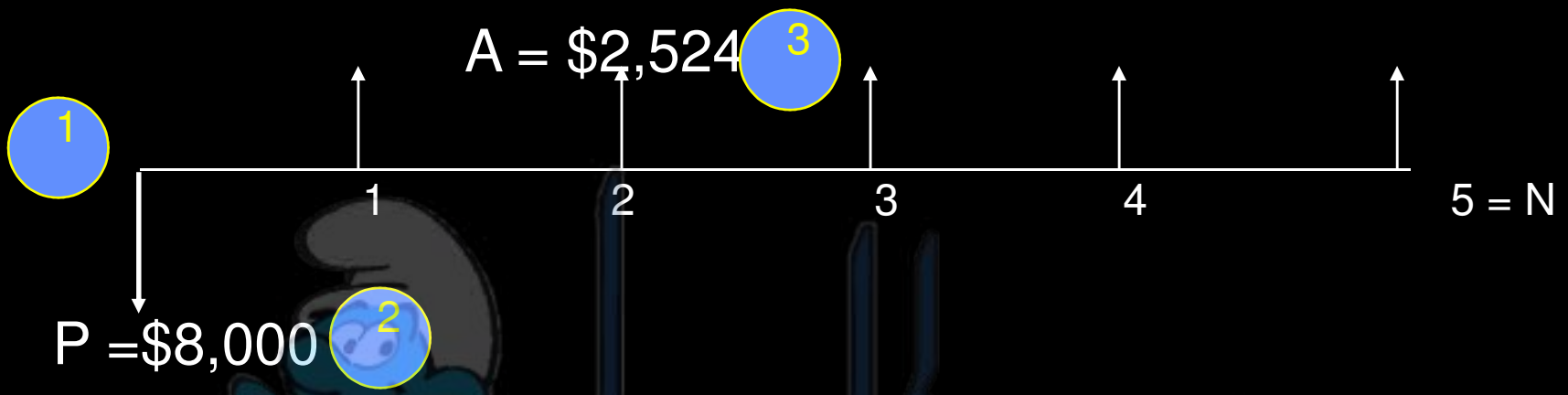
CASH FLOW DIAGRAM NOTATION



1 Time scale with progression of time moving from left to right; the numbers represent time periods (e.g., years, months, quarters, etc...) and may be presented within a time interval or at the end of a time interval.

2 Present expense (cash outflow) of $8,000 for lender.

CASH FLOW DIAGRAM NOTATION

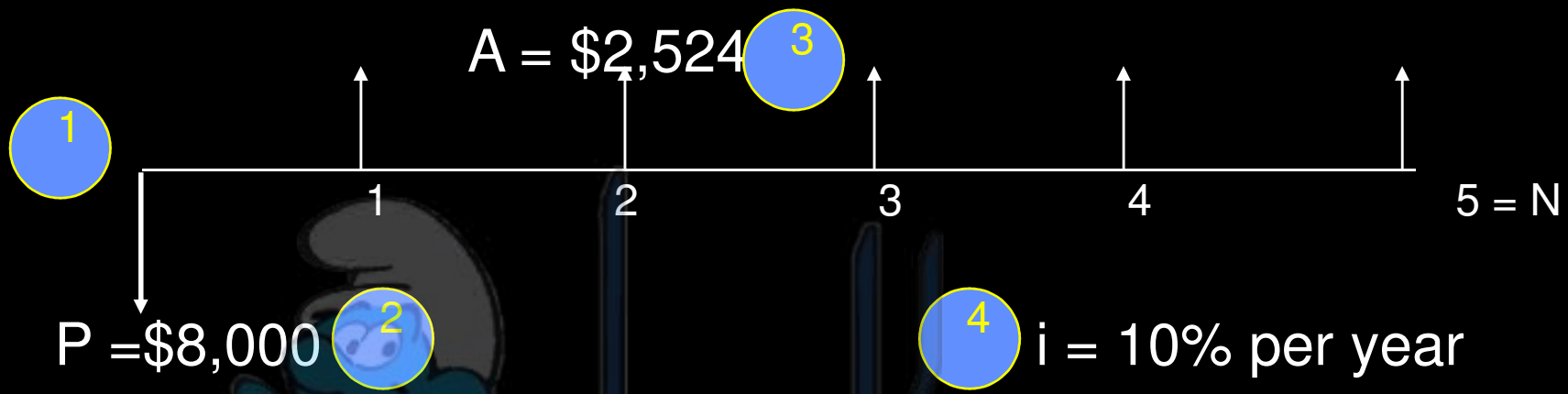


1. Time scale with progression of time moving from left to right; the numbers represent time periods (e.g., years, months, quarters, etc...) and may be presented within a time interval or at the end of a time interval.

2. Present expense (cash outflow) of $8,000 for lender.

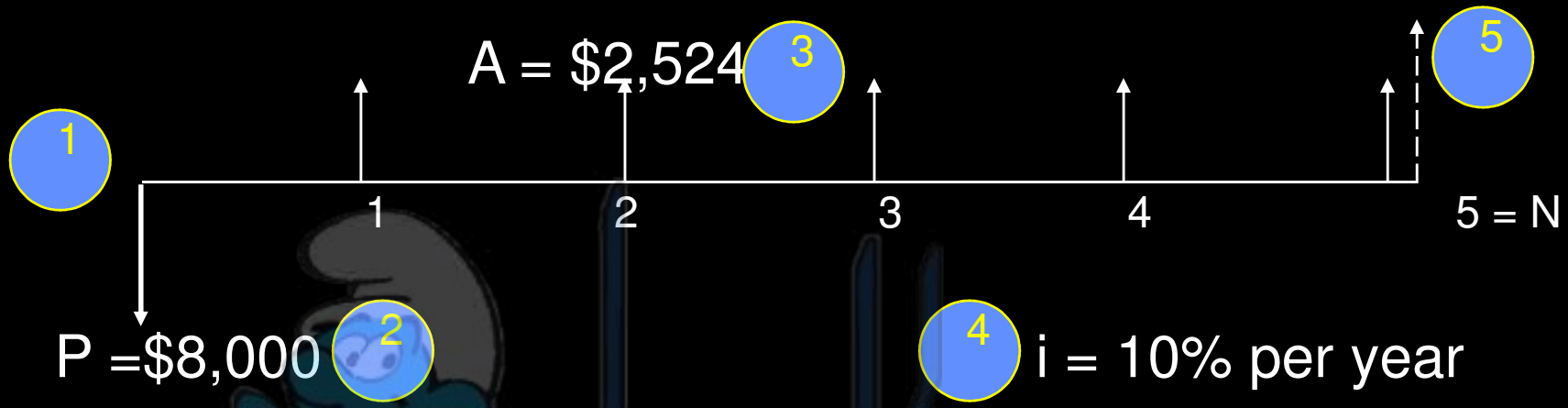
3. Annual income (cash inflow) of $2,524 for lender.

CASH FLOW DIAGRAM NOTATION



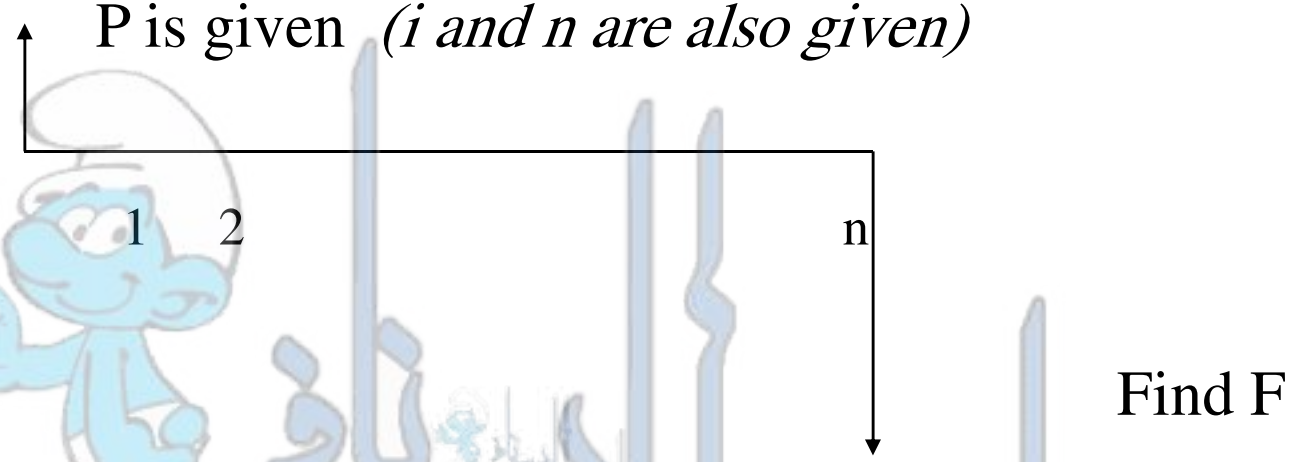
1. Time scale with progression of time moving from left to right; the numbers represent time periods (e.g., years, months, quarters, etc...) and may be presented within a time interval or at the end of a time interval.
2. Present expense (cash outflow) of $8,000 for lender.
3. Annual income (cash inflow) of $2,524 for lender.
4. Interest rate of loan.

CASH FLOW DIAGRAM NOTATION



1. Time scale with progression of time moving from left to right; the numbers represent time periods (e.g., years, months, quarters, etc...) and may be presented within a time interval or at the end of a time interval.
2. Present expense (cash outflow) of $8,000 for lender.
3. Annual income (cash inflow) of $2,524 for lender.
4. Interest rate of loan.
5. Dashed-arrow line indicates amount to be determined.

Single Payment Factors: (F/P, i ,n)



Recall that $F = P(1 + i)^n$

Given P, to find F, the conversion factor is $(1+i)^n$

$F = P\,(F/P, i, n)$

where $(F/P, i, n) = (1+ i)^n$

(F/P, i, n) is tabulated for different i and n

FV(i%,n,,P) in Excel

تم التحميل من شبكة المهندس المسلم
www.muslimengineer.info

Single Payment Factors: (F/P, i ,n)

If $1,000 were deposited in a savings a/c, what would be the ac/ balance in two years if the bank paid 4% interest per year compounded annually?

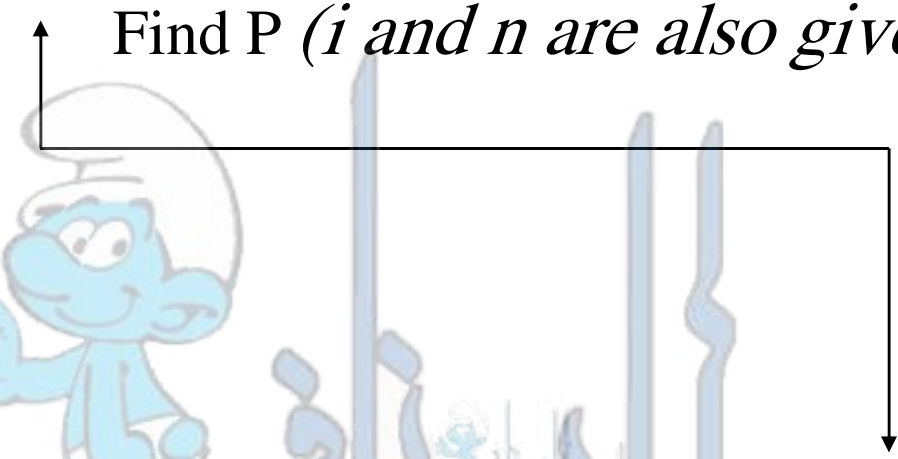
Solution: P = $1,000, n = 2, i = 4%, F = ?

$F = P(1+i)^n = \$1{,}000\ (1 + 0.04)^2 = \$1{,}081.60$, *or*

the factor (F/P, 4%, 2) is 1.0818

F = P (F/P, 4%, 2) = $1,000 (1.0816) = 1,081.80

Single Payment Factor: (P/F, i ,n)



$F = P(1 + i)^n$ or $P = F/(1 + i)^n$

So ***$1/(1+i)^n$ is the P/F conversion factor***

$$P = F\,(P/F, i, n)$$

$$where\ (P/F, i, n) = 1/(1+i)^n$$

(P/F, i, n) is tabulated for different i

PV(i%,n,,F) in Excel

تم التحميل من شبكة المهندس المسلم
www.muslimengineer.info

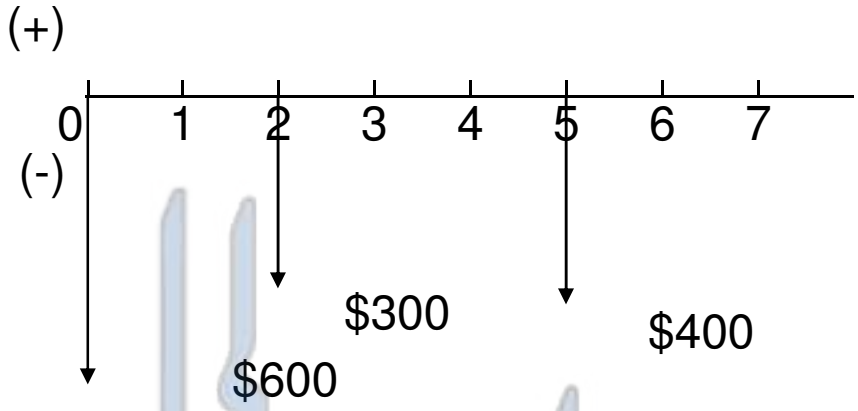
Single Payment Factor: (P/F, i ,n

If you wished to have $1,082 in a savings account at the end of two years and 4% interest was paid annually, how much should you put into the savings account now?

<u>Solution</u>: P = ?, n = 2, i = 4%, F = $1,082

$P = F/(1+i)^n = \$1,082/(1 + 0.04)^2 = \$1,000.37$, ***or***

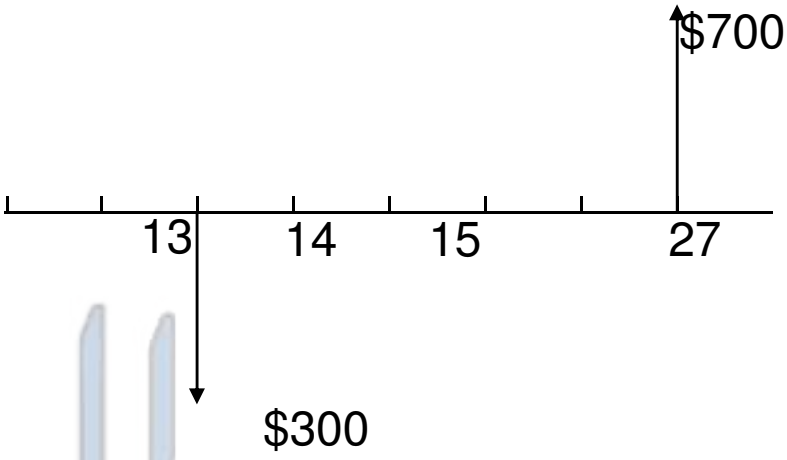
P = F (P/F, 4%, 2) = $1,000.37



Find the future worth of this cash flow series 10 Y at an interest rate of 5% per year.

Solution: F_{10} = - 600 (F/P,5%,10) – 300 (F/P,5%,8) – 400 (F/P,5%,5) = -1931.08

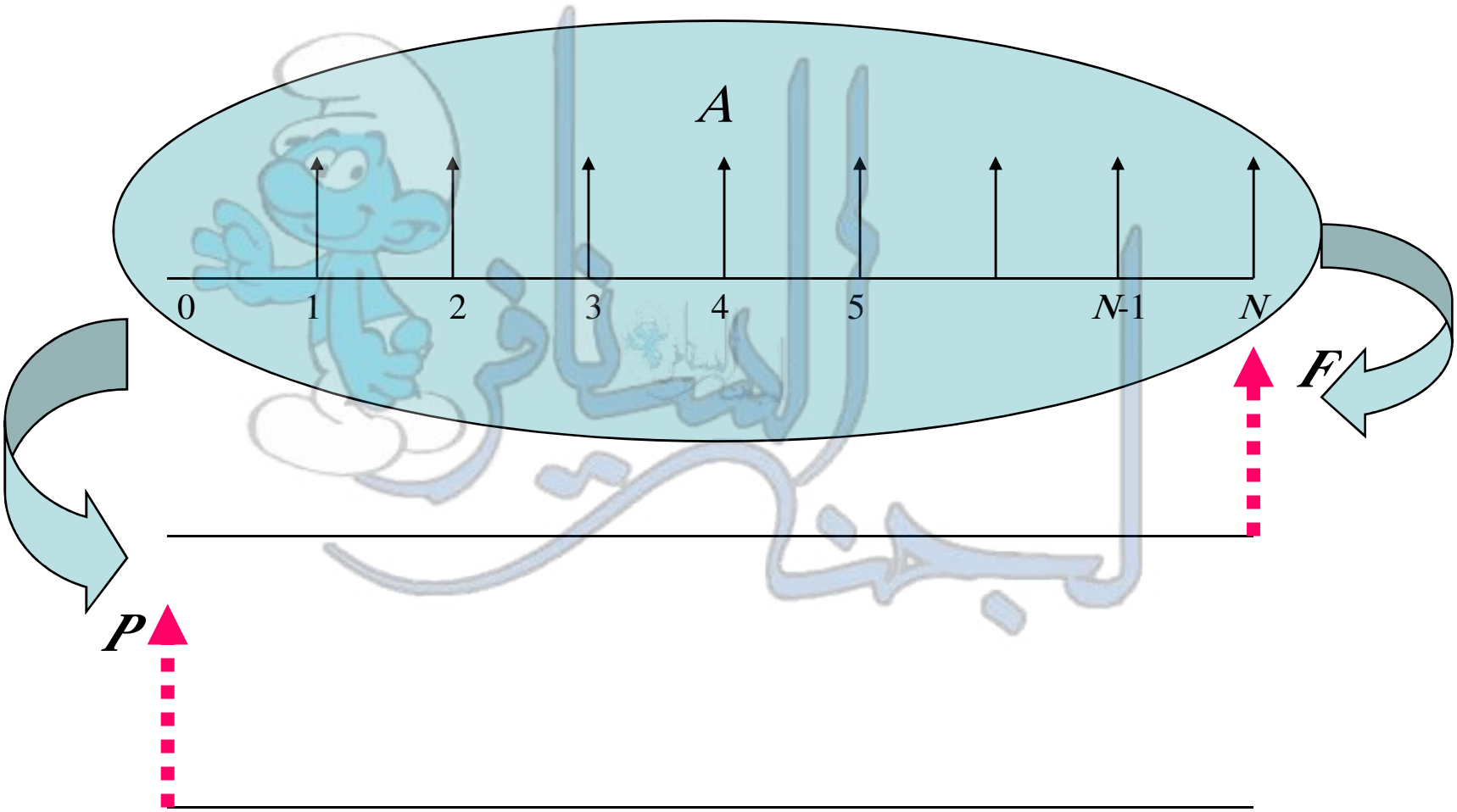
تم التحميل من شبكة المهندس المسلم
www.muslimengineer.info



Find the equivalent single payment of this cash flow series 15 Y at an interest rate of 5% per year.

Solution: F_{15} = -300(F/P,5%,2) +700(P/F,5%,12) = 59.04

Equal Payment Series



تم التحميل من شبكة المهندس المسلم
www.muslimengineer.info

Uniform Series Present Worth Factor (P/A, i , n)

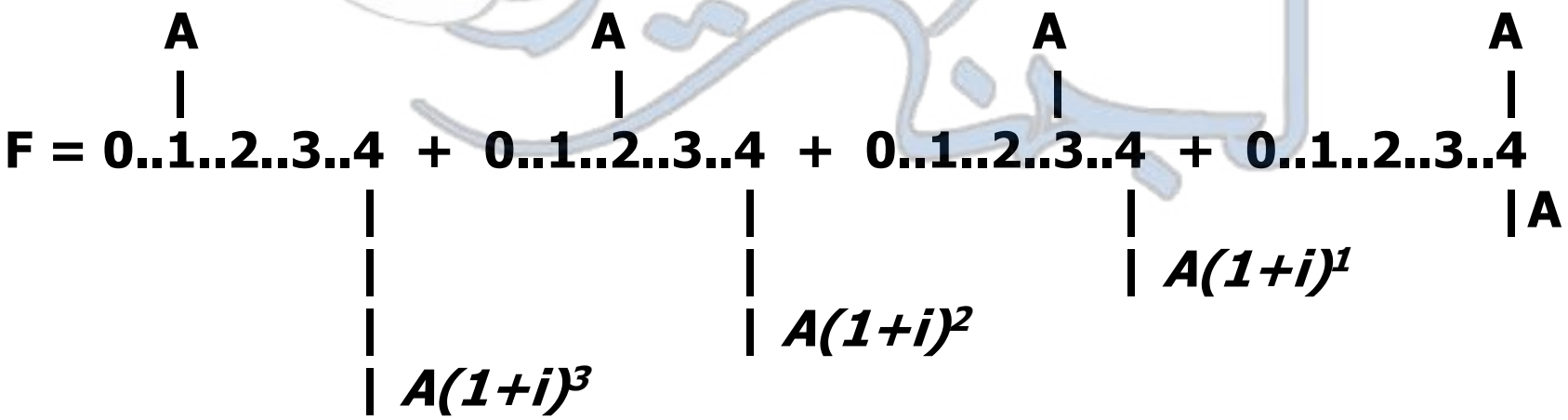


- Objective: Find P, given A

A uniform series of payments or receipts represents:

A collection of end-of-period cash payments or receipts arranged in a uniform series and continuing for n periods. Such a series is equivalent to P or F at interest rate i, given the constant cash payment (or receipt) designated as A (based on the term "annuity", a regular payment).

Consider a 4-yr period:



تم التحميل من شبكة المهندس المسلم
www.muslimengineer.info

Uniform series (contin.)

$F = A(1+i)^3 + A(1+i)^2 + A(1+i)^1 + A$ Now, multiply by (1+i)

$(1+i)\,F = A(1+i)^4 + A(1+i)^3 + A(1+i)^2 + A(1+i)$

$-\ F = A(1+i)^3 + A(1+i)^2 + A(1+i)^1 + A$ Solve for the difference

$i\,F = A(1+i)^4 - A$

$= A[(1+i)^4 - 1]$ Thus, $F = A[(1+i)^4 - 1] / i$

In general,

$$F = A\left[\frac{(1+i)^n - 1}{i}\right]$$

\ uniform series compound-amount factor

Uniform series (contin.)

If we turn this around and solve for A, we obtain:

$$A = F\left[\frac{i}{(1+i)^n - 1}\right]$$

\ uniform series

sinking-fund factor

Example: Set up a uniform-payment investment (college fund) with the goal of having $80,000 after 20 years, invested at 6% compounded annually. What is the required annual payment?

A = $80,000(.06)/[1.06^{20} –1] = $80,000(.06/2.207135) = $2174.77

OR

A = F (A/F, 6%, 20) = $80,000(.0272) = $2176

(~ $182/mo.)

تم التحميل من شبكة المهندس المسلم
www.muslimengineer.info

Uniform series (contin.)

If we take the uniform-series compounding equation and replace F with the single-payment compounding expression, we obtain:

$$P(1+i)^n = A\left[\frac{(1+i)^n - 1}{i}\right]$$

uniform series present-worth factor

$$P = A\left[\frac{(1+i)^n - 1}{i(1+i)^n}\right]$$

This expression is used to calculate the present worth, given the regular annuity payment.

Example:

Our consulting firm would like to purchase a used testing machine from an independent testing/inspection lab, and we make two offers: 1) a lump-sum of $40,000 or 2) monthly payments of $1200 over 3 years at a 6% annual interest rate. Which option do you think the testing lab would prefer, assuming it has to replace the sold machine?

$$P_{loan} = \$1200\,[P/A, 0.5\%, 36] = \$1200(32.871) = \$39{,}445$$

The lab would prefer the $40000 payment now, because it is greater than the present worth of the proposed loan terms.

Note: $F_{loan} = \$1200[F/A, 0.5\%, 36] = \$1200(39.336) = \$47{,}203$

تم التحميل من شبكة المهندس المسلم
www.muslimengineer.info

Uniform series (contin.)

If we instead solve for A, we obtain:

$$A = P\left[\frac{i(1+i)^n}{(1+i)^n - 1}\right]$$

uniform series capital-recovery factor

This expression is used to calculate the regular annuity payment, given the present worth.

Standard Factor Notation

To Find	Given	Factor	Equation
P	F	(P/F, i, n)	P = F*(P/F, i, n)
F	P	(F/P, i, n)	F = P*(F/P, i, n)
P	A	(P/A, i, n)	P = A*(P/A, i, n)
A	P	(A/P, i, n)	A = P*(A/P, i, n)
A	F	(A/F, i, n)	A = F*(A/F, i, n)
F	A	(F/A, i, n)	F = A*(F/A, i, n)

Practice deriving these factor formulas directly using geometric sum identity

تم التحميل من شبكة المهندس المسلم
www.muslimengineer.info

Uniform Series Compound Amount Tables

n	(F/A, 4%, n)	(F/A, 5%, n)	(F/A, 6%, n)	(F/A, 7%, n)	(F/A, 8%, n)	(F/A, 9%, n)	(F/A, 10%, n)
1	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2	2.040	2.050	2.060	2.070	2.080	2.090	2.100
3	3.122	3.153	3.184	3.215	3.246	3.278	3.310
4	4.246	4.310	4.375	4.440	4.506	4.573	4.641
5	5.416	5.526	5.637	5.751	5.867	5.985	6.105
6	6.633	6.802	6.975	7.153	7.336	7.523	7.716
7	7.898	8.142	8.394	8.654	8.923	9.20(	
8	9.214	9.549	9.897	10.260	10.637	11.02	
9	10.583	11.027	11.491	11.978	12.488	13.02	
10	12.006	12.578	13.181	13.816	14.487	15.193	15.937
11	13.486	14.207	14.972	15.784	16.645	17.560	18.531
12	15.026	15.917	16.870	17.888	18.977	20.141	21.384
13	16.627	17.713	18.882	20.141	21.495	22.953	24.523
14	18.292	19.599	21.015	22.550	24.215	26.019	27.975
15	20.024	21.579	23.276	25.129	27.152	29.361	31.772
16	21.825	23.657	25.673	27.888	30.324	33.003	35.950
17	23.698	25.840	28.213	30.840	33.750	36.974	40.545
18	25.645	28.132	30.906	33.999	37.450	41.301	45.599
19	27.671	30.539	33.760	37.379	41.446	46.018	51.159
20	29.778	33.066	36.786	40.995	45.762	51.160	57.275

$$\frac{(1+0.08)^5-1}{0.08}$$

Uniform Series Sinking Fund Tables

n	(A/F, 4%, n)	(A/F, 5%, n)	(A/F, 6%, n)	(A/F, 7%, n)	(A/F, 8%, n)	(A/F, 9%, n)	(A/F, 10%, n)
1	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
2	0.4902	0.4878	0.4854	0.4831	0.4808	0.4785	0.4762
3	0.3203	0.3172	0.3141	0.3111	0.3080	0.3051	0.3021
4	0.2355	0.2320	0.2286	0.2252	0.2219	0.2187	0.2155
5	0.1846	0.1810	0.1774	0.1739	0.1705	0.1671	0.1638
6	0.1508	0.1470	0.1434	0.1398	0.1363	0.1329	0.1296
7	0.1266	0.1228	0.1191	0.1156	0.1121	0.1	4
8	0.1085	0.1047	0.1010	0.0975	0.0940	0.0	4
9	0.0945	0.0907	0.0870	0.0835	0.0801	0.0	5
10	0.0833	0.0795	0.0759	0.0724	0.0690	0.0658	0.0627
11	0.0741	0.0704	0.0668	0.0634	0.0601	0.0569	0.0540
12	0.0666	0.0628	0.0593	0.0559	0.0527	0.0497	0.0468
13	0.0601	0.0565	0.0530	0.0497	0.0465	0.0436	0.0408
14	0.0547	0.0510	0.0476	0.0443	0.0413	0.0384	0.0357
15	0.0499	0.0463	0.0430	0.0398	0.0368	0.0341	0.0315
16	0.0458	0.0423	0.0390	0.0359	0.0330	0.0303	0.0278
17	0.0422	0.0387	0.0354	0.0324	0.0296	0.0270	0.0247
18	0.0390	0.0355	0.0324	0.0294	0.0267	0.0242	0.0219
19	0.0361	0.0327	0.0296	0.0268	0.0241	0.0217	0.0195
20	0.0336	0.0302	0.0272	0.0244	0.0219	0.0195	0.0175

$$\frac{0.08}{(1+0.08)^5-1}$$

تم التحميل من شبكة المهندس المسلم
www.muslimengineer.info

Uniform Series Capital Recovery Tables

n	(A/P, 4%, n)	(A/P, 5%, n)	(A/P, 6%, n)	(A/P, 7%, n)	(A/P, 8%, n)	(A/P, 9%, n)	(A/P, 10%, n)
1	1.0400	1.0500	1.0600	1.0700	1.0800	1.0900	1.1000
2	0.5302	0.5378	0.5454	0.5531	0.5608	0.5685	0.5762
3	0.3603	0.3672	0.3741	0.3811	0.3880	0.3951	0.4021
4	0.2755	0.2820	0.2886	0.2952	0.3019	0.3087	0.3155
5	0.2246	0.2310	0.2374	0.2439	0.2505	0.2571	0.2638
6	0.1908	0.1970	0.2034	0.2098	0.2163	0.2229	0.2296
7	0.1666	0.1728	0.1791	0.1856	0.1921	0	
8	0.1485	0.1547	0.1610	0.1675	0.1740	0	
9	0.1345	0.1407	0.1470	0.1535	0.1601	0	
10	0.1233	0.1295	0.1359	0.1424	0.1490	0.1558	0.1627
11	0.1141	0.1204	0.1268	0.1334	0.1401	0.1469	0.1540
12	0.1066	0.1128	0.1193	0.1259	0.1327	0.1397	0.1468
13	0.1001	0.1065	0.1130	0.1197	0.1265	0.1336	0.1408
14	0.0947	0.1010	0.1076	0.1143	0.1213	0.1284	0.1357
15	0.0899	0.0963	0.1030	0.1098	0.1168	0.1241	0.1315
16	0.0858	0.0923	0.0990	0.1059	0.1130	0.1203	0.1278
17	0.0822	0.0887	0.0954	0.1024	0.1096	0.1170	0.1247
18	0.0790	0.0855	0.0924	0.0994	0.1067	0.1142	0.1219
19	0.0761	0.0827	0.0896	0.0968	0.1041	0.1117	0.1195
20	0.0736	0.0802	0.0872	0.0944	0.1019	0.1095	0.1175

$$\frac{0.08\times(1+0.08)^5}{(1+0.08)^5-1}$$

Uniform Series Present Worth Tables

n	(P/A, 4%, n)	(P/A, 5%, n)	(P/A, 6%, n)	(P/A, 7%, n)	(P/A, 8%, n)	(P/A, 9%, n)	(P/A, 10%, n)
1	0.962	0.952	0.943	0.935	0.926	0.917	0.909
2	1.886	1.859	1.833	1.808	1.783	1.759	1.736
3	2.775	2.723	2.673	2.624	2.577	2.531	2.487
4	3.630	3.546	3.465	3.387	3.312	3.240	3.170
5	4.452	4.329	4.212	4.100	3.993	3.890	3.791
6	5.242	5.076	4.917	4.767	4.623	4.486	4.355
7	6.002	5.786	5.582	5.389	5.206	5	
8	6.733	6.463	6.210	5.971	5.747	5	
9	7.435	7.108	6.802	6.515	6.247	5	
10	8.111	7.722	7.360	7.024	6.710	6.418	6.145
11	8.760	8.306	7.887	7.499	7.139	6.805	6.495
12	9.385	8.863	8.384	7.943	7.536	7.161	6.814
13	9.986	9.394	8.853	8.358	7.904	7.487	7.103
14	10.563	9.899	9.295	8.745	8.244	7.786	7.367
15	11.118	10.380	9.712	9.108	8.559	8.061	7.606
16	11.652	10.838	10.106	9.447	8.851	8.313	7.824
17	12.166	11.274	10.477	9.763	9.122	8.544	8.022
18	12.659	11.690	10.828	10.059	9.372	8.756	8.201
19	13.134	12.085	11.158	10.336	9.604	8.950	8.365
20	13.590	12.462	11.470	10.594	9.818	9.129	8.514

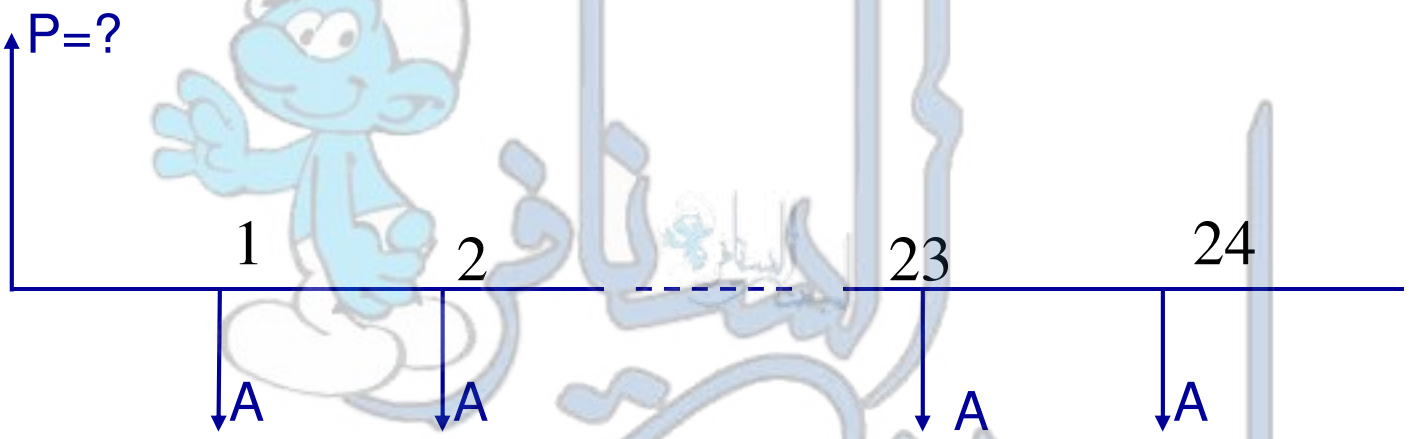
$$\frac{(1+0.08)^5-1}{0.08\times(1+0.08)^5}$$

تم التحميل من شبكة المهندس المسلم
www.muslimengineer.info

Example:

Tom deposits $500 in his saving account at the end of each year for 24 years and the bank pays 6% interest rate per year, compounded yearly. What are the present worth and future worth of this yearly investment.

A = \$500, i = 6%, n = 24 $P = A[\frac{(1+i)^n - 1}{(1+i)^n i}]$

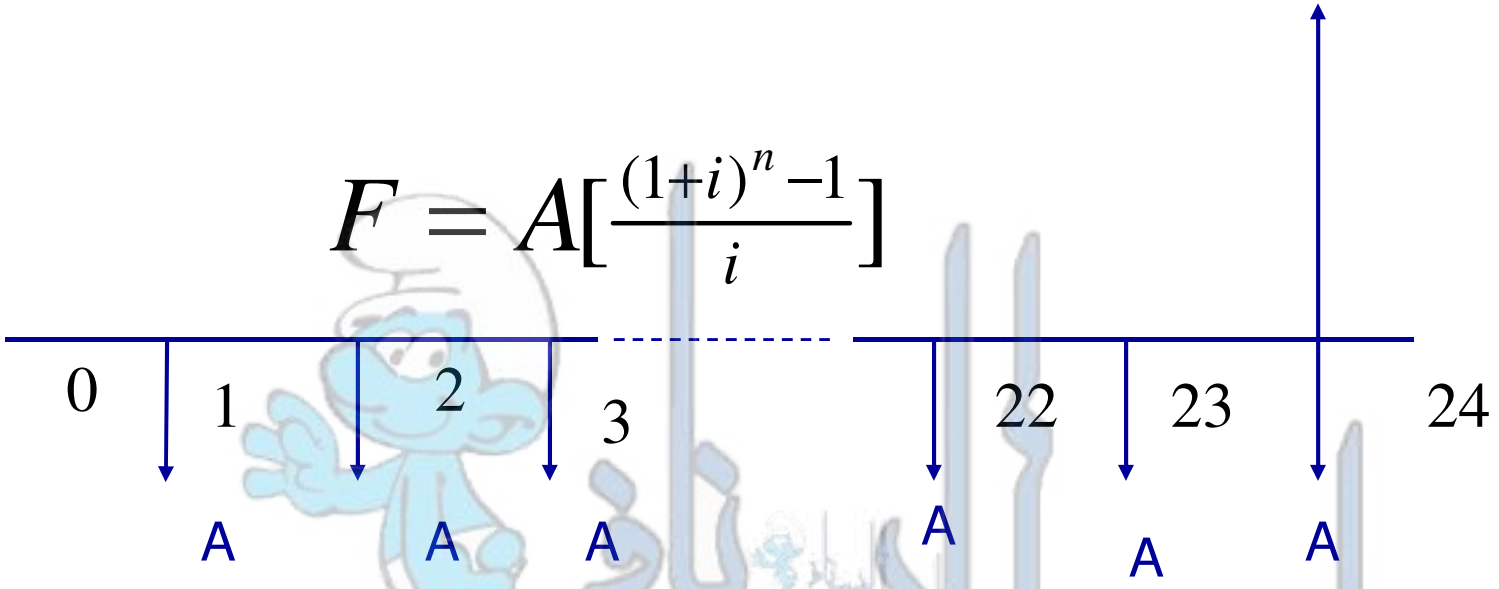


=\$500[(1+0.06)^{24}-1]/[(1+0.06)^{24}(0.06)] = \$6275.18

= 500(P/A, 6%, 24) = 500*12.5504=\$6275.20

تم التحميل من شبكة المهندس المسلم
www.muslimengineer.info

$$F = A[\frac{(1+i)^n - 1}{i}]$$



=$500[(1+0.06)^{24}-1]/0.06

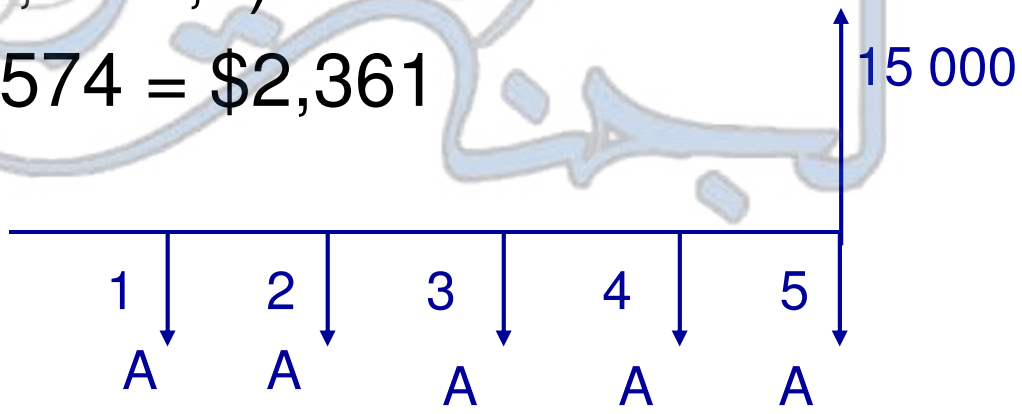
=500(F/A,6%,24) = $500 * 50.815

= $ 25,407.79

Bill wants to make deposits each year for five years to buy the $15,000 car. His first payment will be one year from today. How big must the deposits be if the interest rate is 12% per year?

A = 15000(A/F,12%,5)

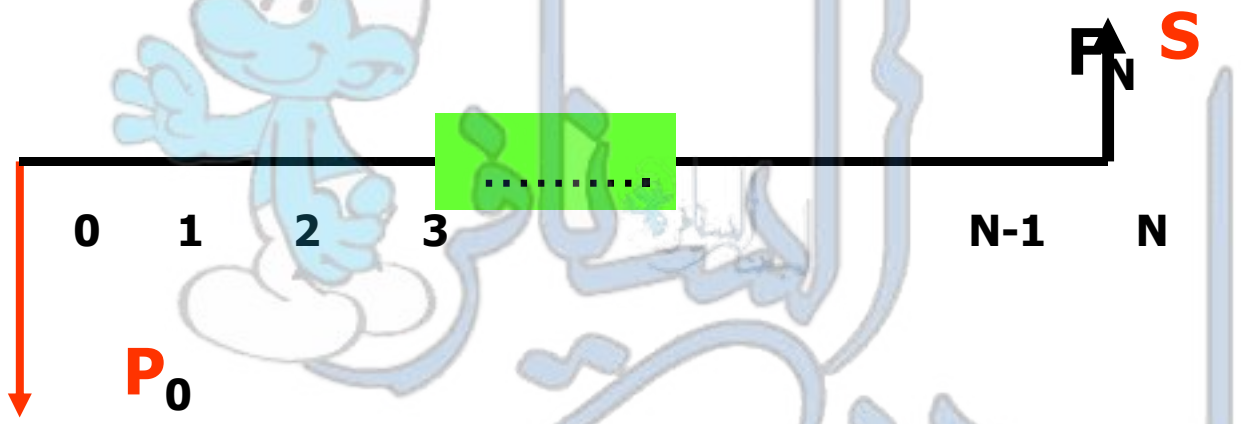
= 15000*0.1574 = $2,361



تم التحميل من شبكة المهندس المسلم
www.muslimengineer.info

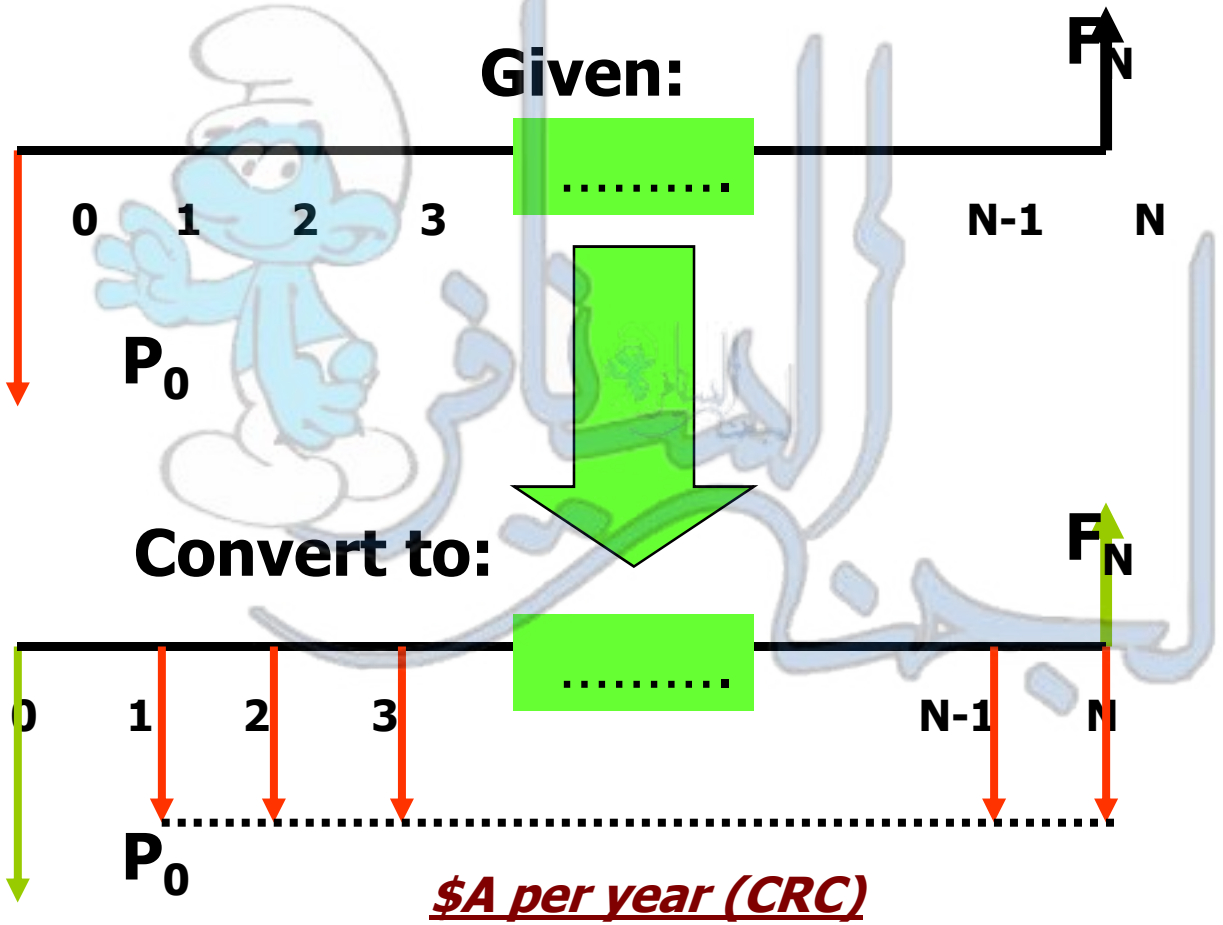
(CR) Capital Cost Recovery

القيمة السنوية المكافئة للمشروع المعطي = CR



استعادة رأس المال هو عبارة عن القيمة السنوية المكافئة للكلفة الاولية و القيمة السنوية لقيمة الخلاص

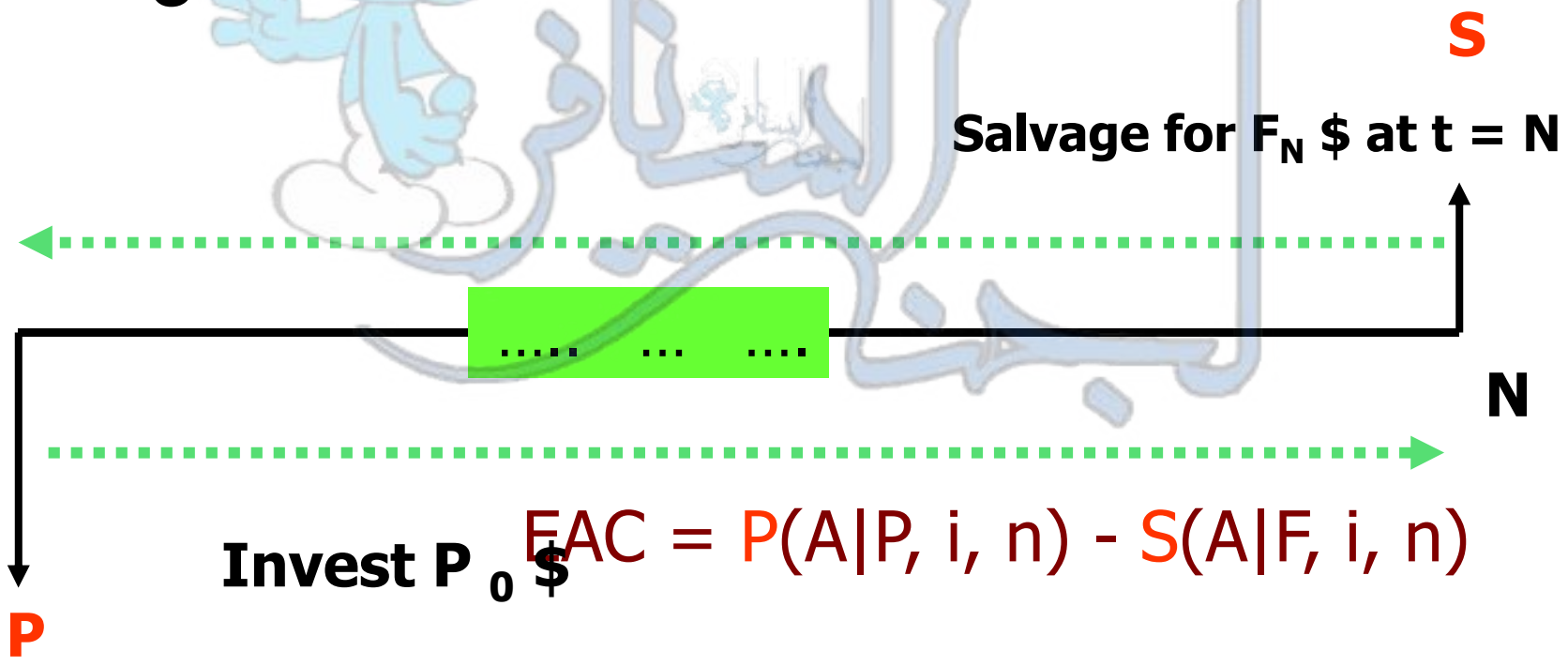
كلفة استعادة رأس المال CR$_S$



تم التحميل من شبكة المهندس المسلم
www.muslimengineer.info

كلفة استعادة رأس المال

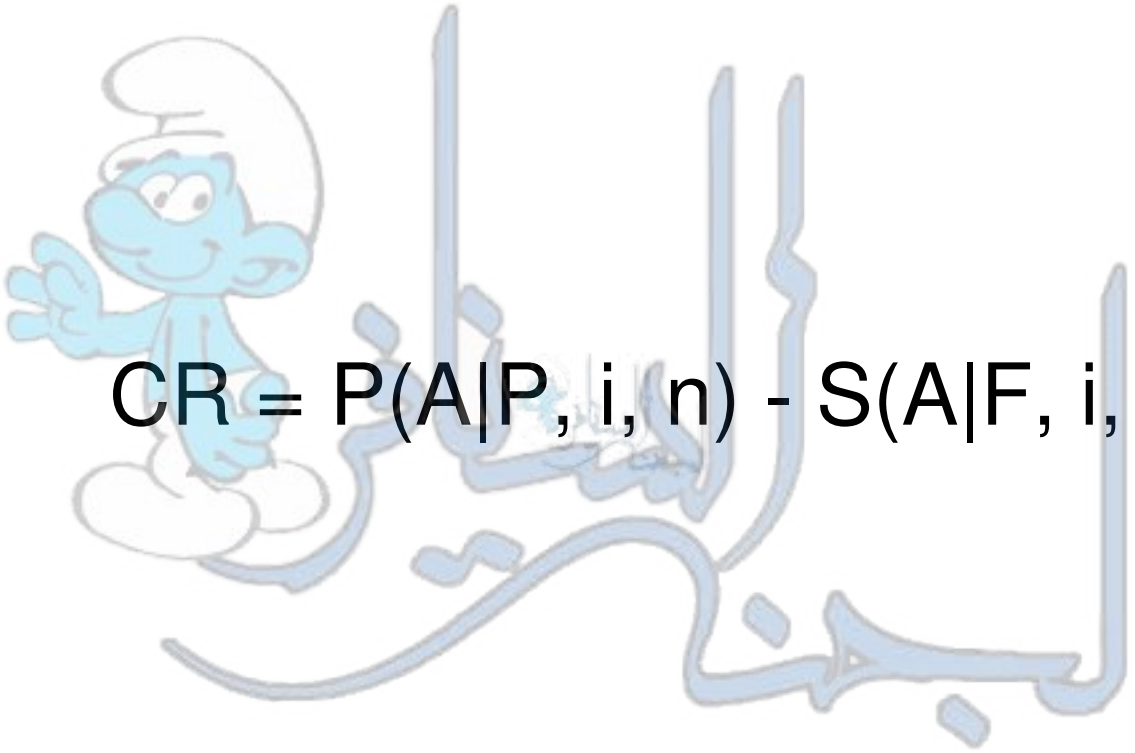
نقوم بحساب هذه الكلفة للايرادات مع قيم الخلاص



$$EAC = P(A|P, i, n) - S(A|F, i, n)$$

Cost = "+" and SV = "-" by convention

كلفة استعادة رأس المال



$$CR = P(A|P, i, n) - S(A|F, i, n)$$

تم التحميل من شبكة المهندس المسلم
www.muslimengineer.info

كلفة استعادة رأس المال

طريقة ثانية – اطرح قيمة الخلاص من الكلفة الأصلية واحسب التكلفة السنوية للفرق . أضف للناتج الفائدة التي تعطيها قيمة الخلاص كل سنة ، SV(I) .

$$CR= (P - S) (A|P, i, n) + S(i)$$

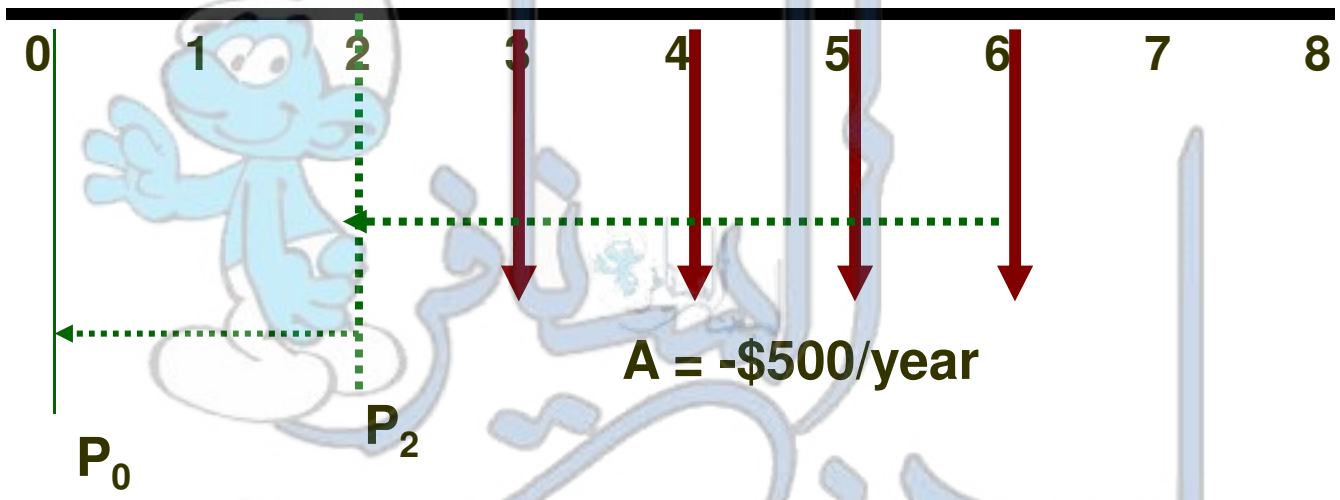
1. Uniform Series that are SHIFTED

- A ***shifted*** series is one whose present worth point in time is NOT $t = 0$.
- Shifted either to the left of t = "0" or to the right of t = "0".
- Dealing with a uniform series:
  - **The PW point is always one period to the left of the first series value**
  - No matter where the series falls on the time line.

تم التحميل من شبكة المهندس المسلم
www.muslimengineer.info

Example :Shifted Series

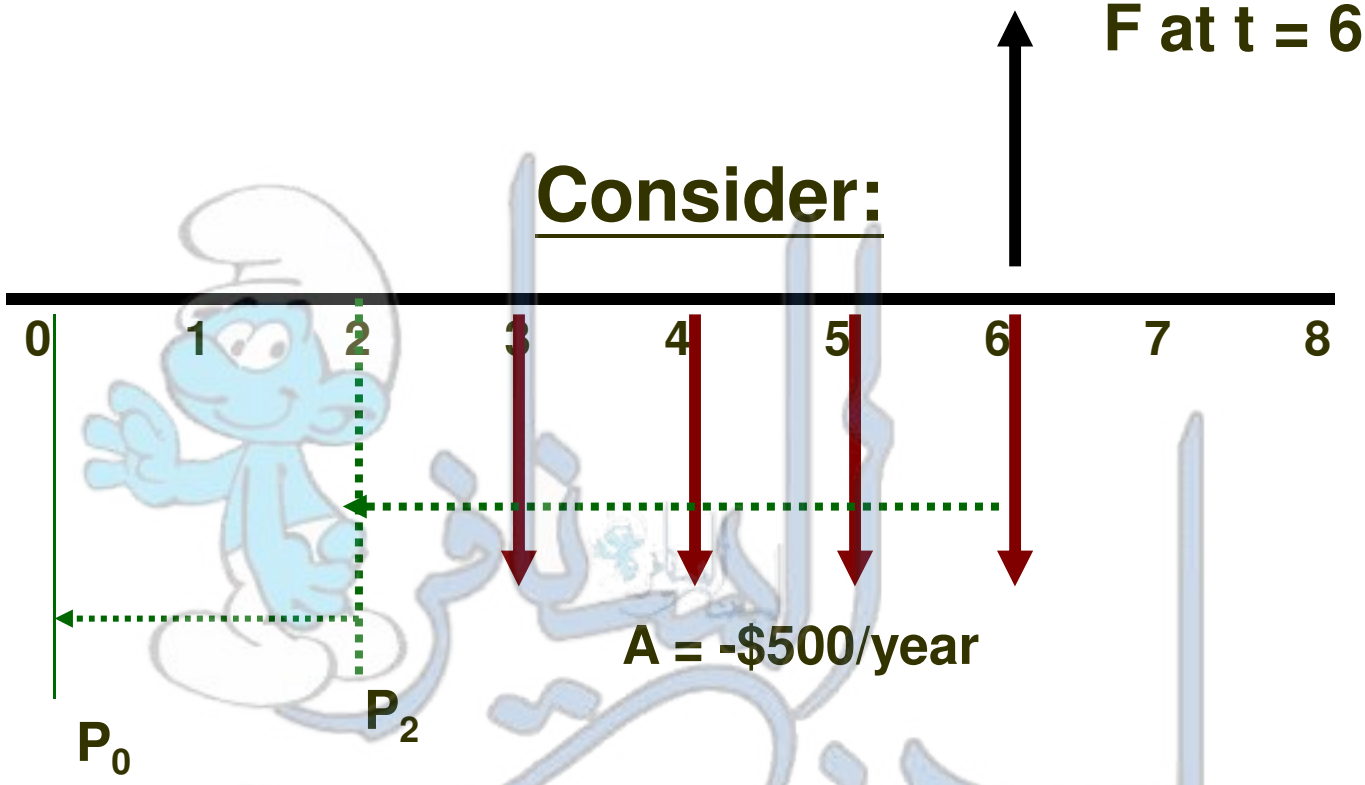
Consider:



P of this series is at t = 2 (P_2 or F_2)

$$P_2 = \$500(P/A,i\%,4),$$

$$P_0 = P_2(P/F,i\%,2)$$



F for this series is at t = 6

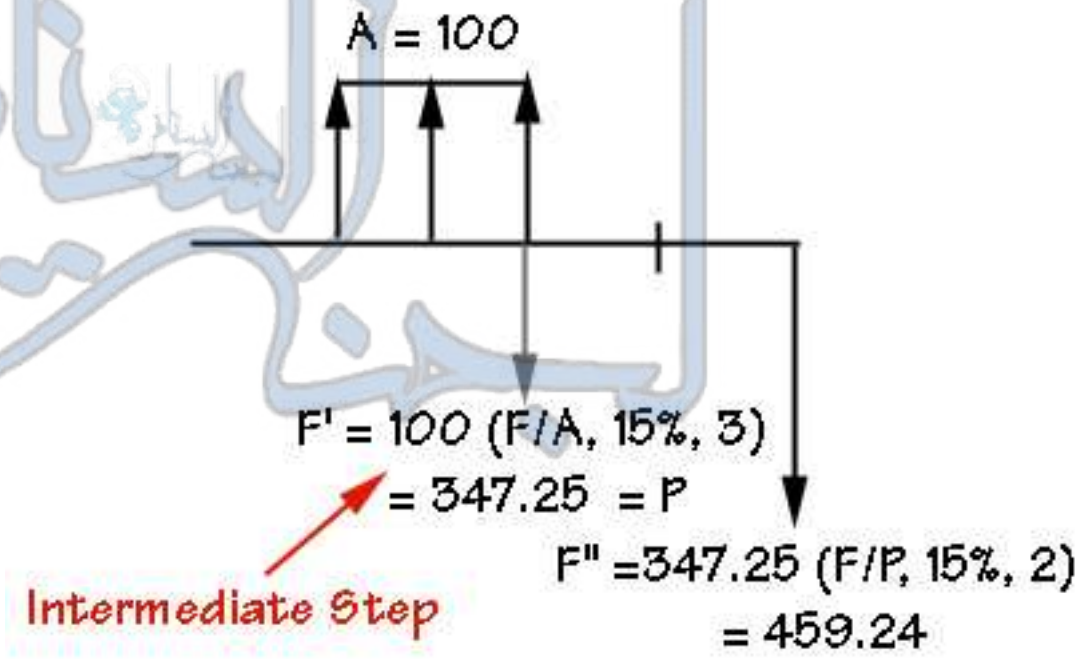
$$F_6 = A(F/A,i\%,4)$$

تم التحميل من شبكة المهندس المسلم
www.muslimengineer.info

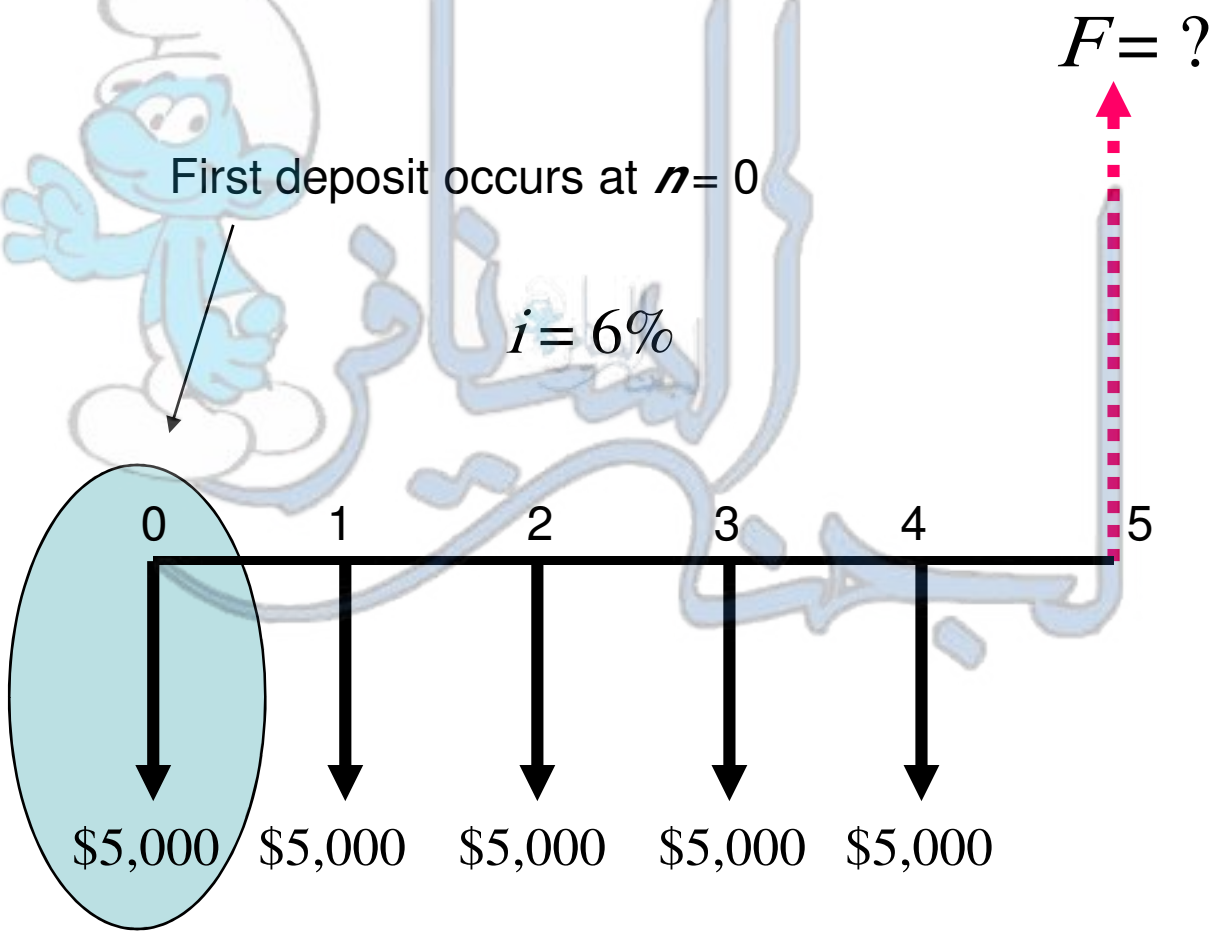
Example

$$F' = \$100(F/A, 15\%, 3) = \$347.25$$
$$F'' = \$347.25(F/P, 15\%, 2) = \$459.24$$

Year	Cash flow
1	$100
2	$100
3	$100
4	$0
5	F



Example : Handling Time Shifts in a Uniform Series

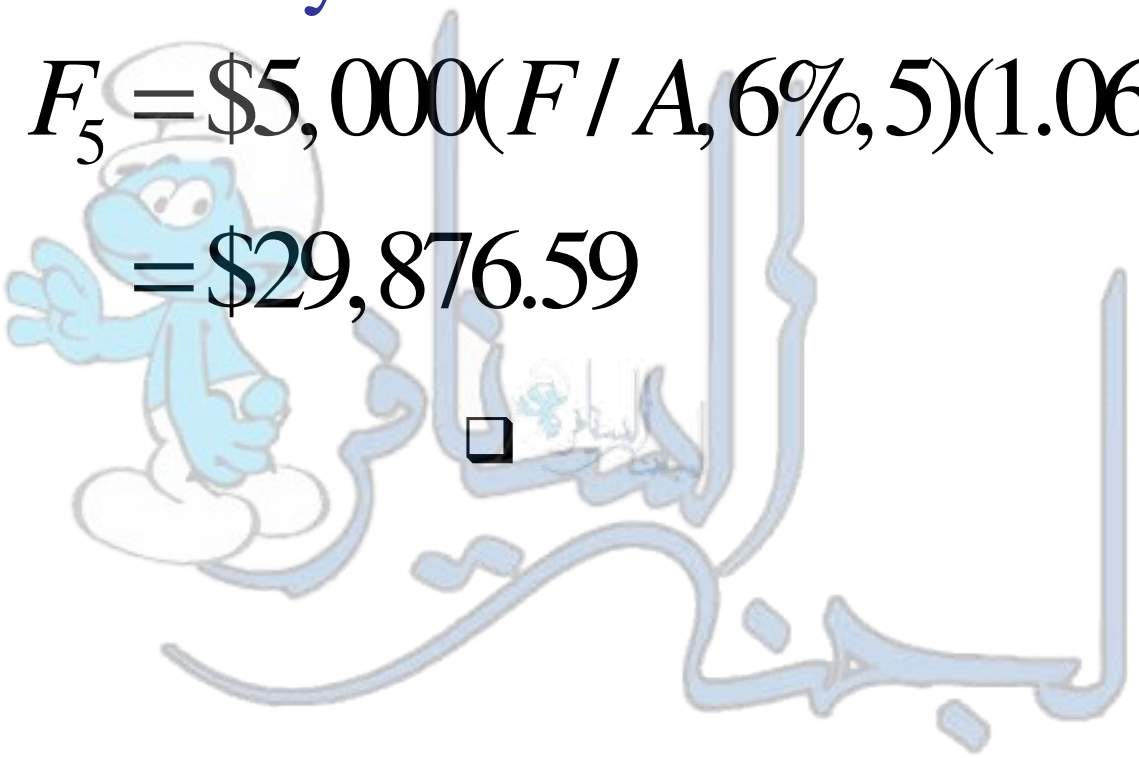


تم التحميل من شبكة المهندس المسلم
www.muslimengineer.info

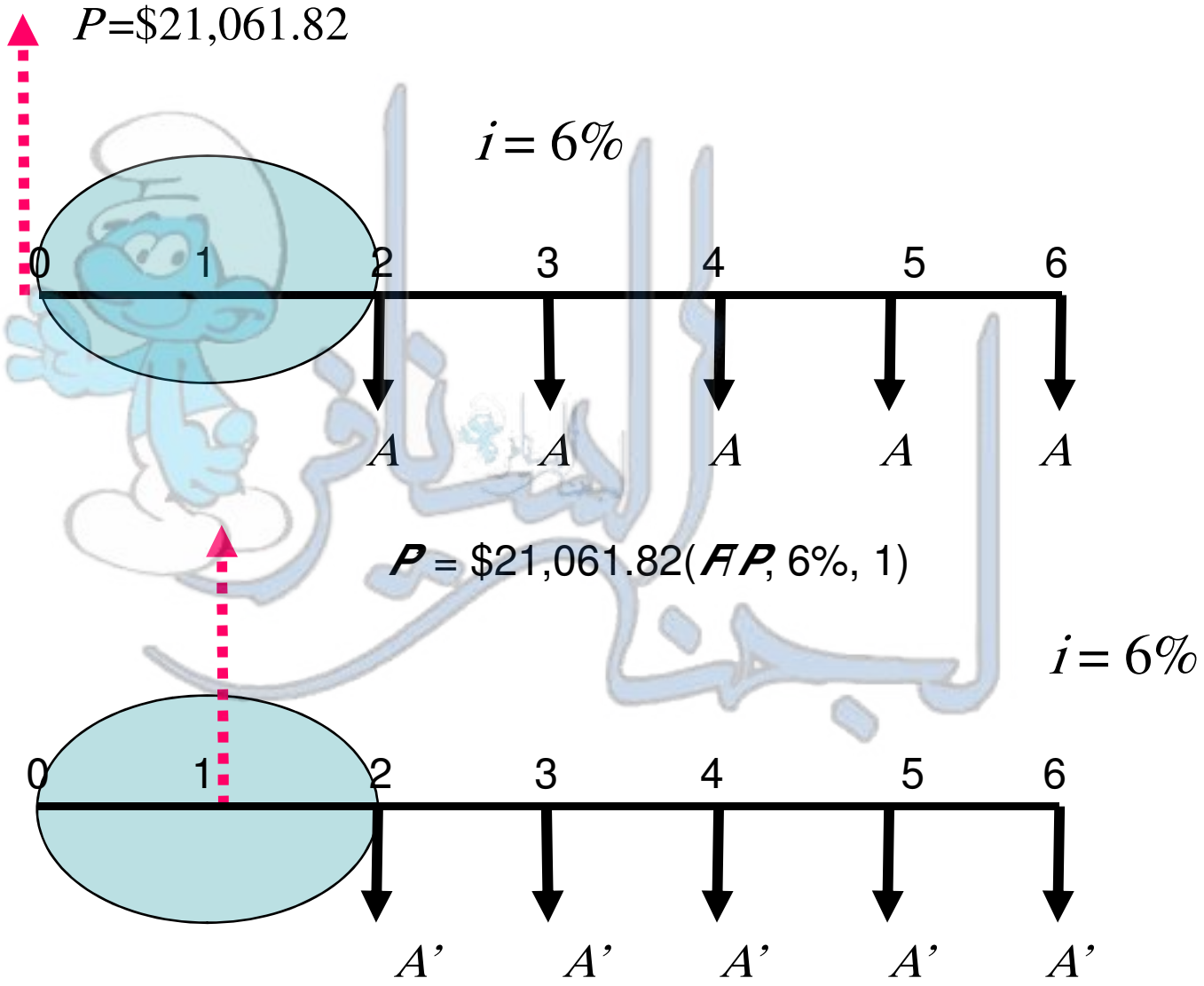
- Annuity Due

$$F_5 = \$5,000(F/A, 6\%, 5)(1.06)$$

$$= \$29,876.59$$



Example : Deferred Loan Repayment Plan



تم التحميل من شبكة المهندس المسلم
www.muslimengineer.info

Two-Step Procedure

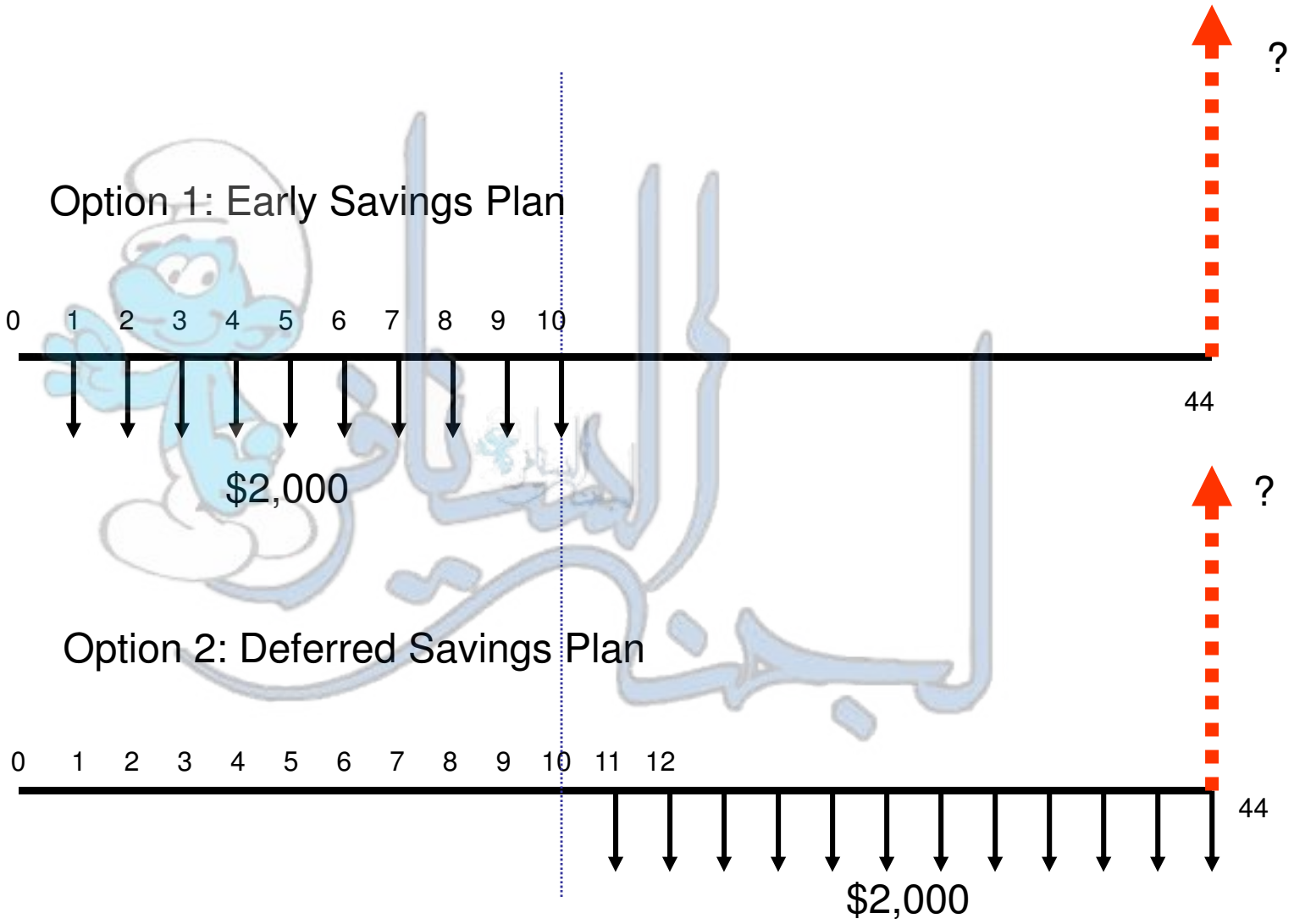
$$P' = \$21,061.82(F/P,6\%,1)$$

$$= \$22,325.53$$

$$A = \$22,325.53(A/P,6\%,5)$$

$$= \$5,300$$

Example : Early Savings Plan – 8% interest

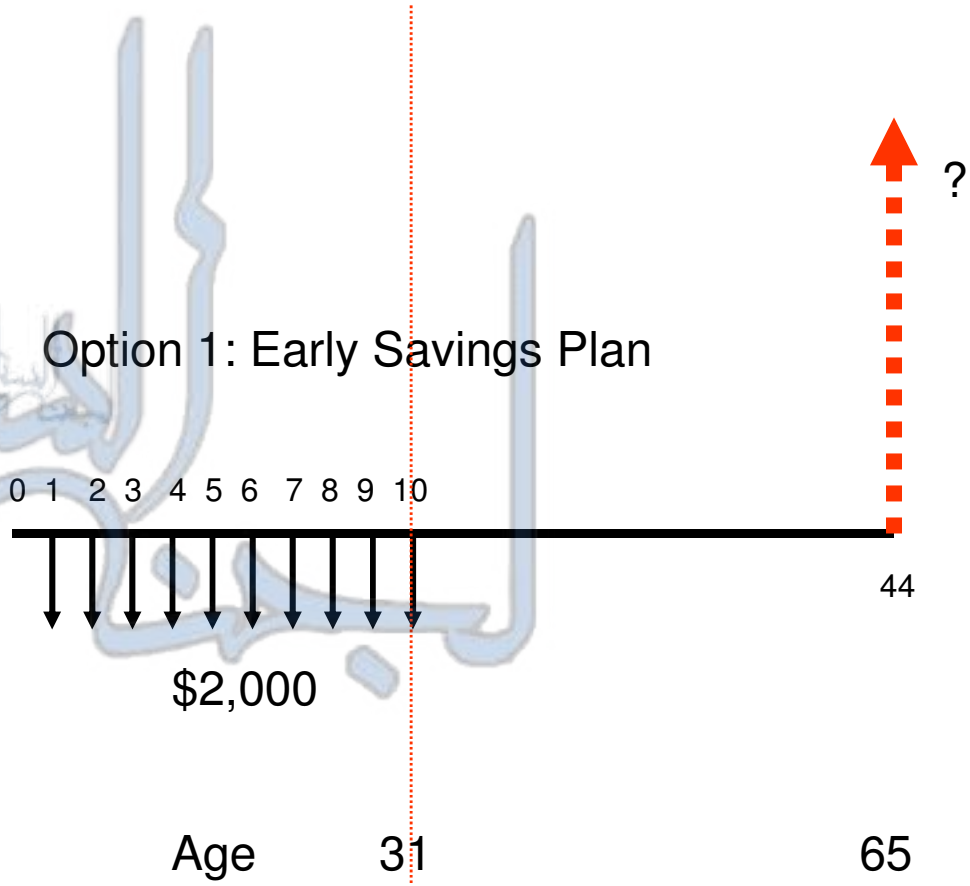


تم التحميل من شبكة المهندس المسلم
www.muslimengineer.info

Option 1 – Early Savings Plan

$$F_{10} = \$2,000(F/A, 8\%, 10)$$
$$= \$28,973$$

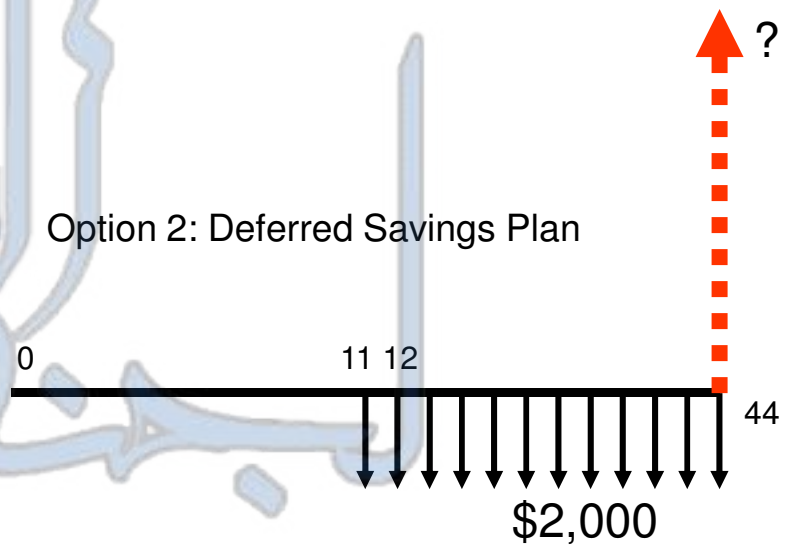
$$F_{44} = \$28,973(F/P, 8\%, 34)$$
$$= \$396,645$$



Option 2: Deferred Savings Plan

$$F_{44} = \$2{,}000(F/A, 8\%, 34)$$

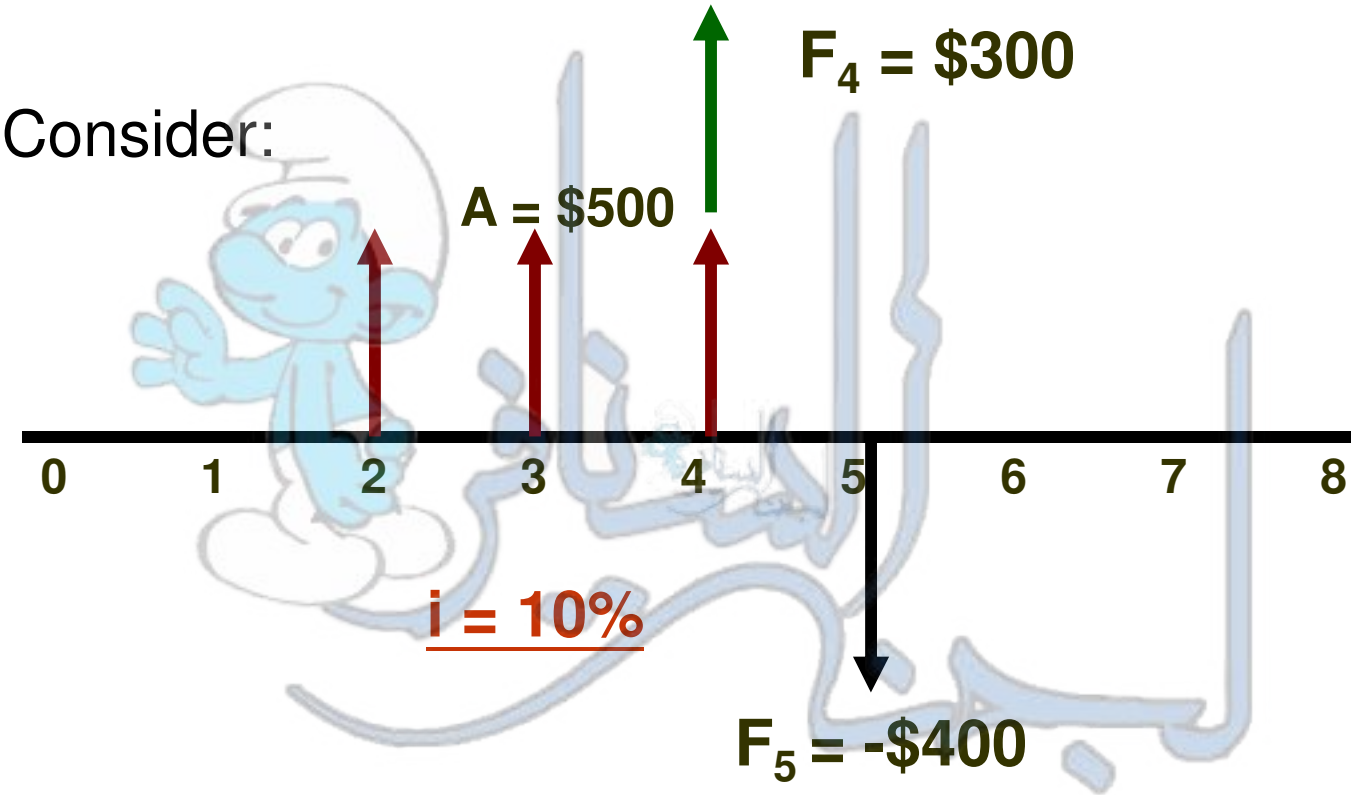
$$= \$317{,}233$$



تم التحميل من شبكة المهندس المسلم
www.muslimengineer.info

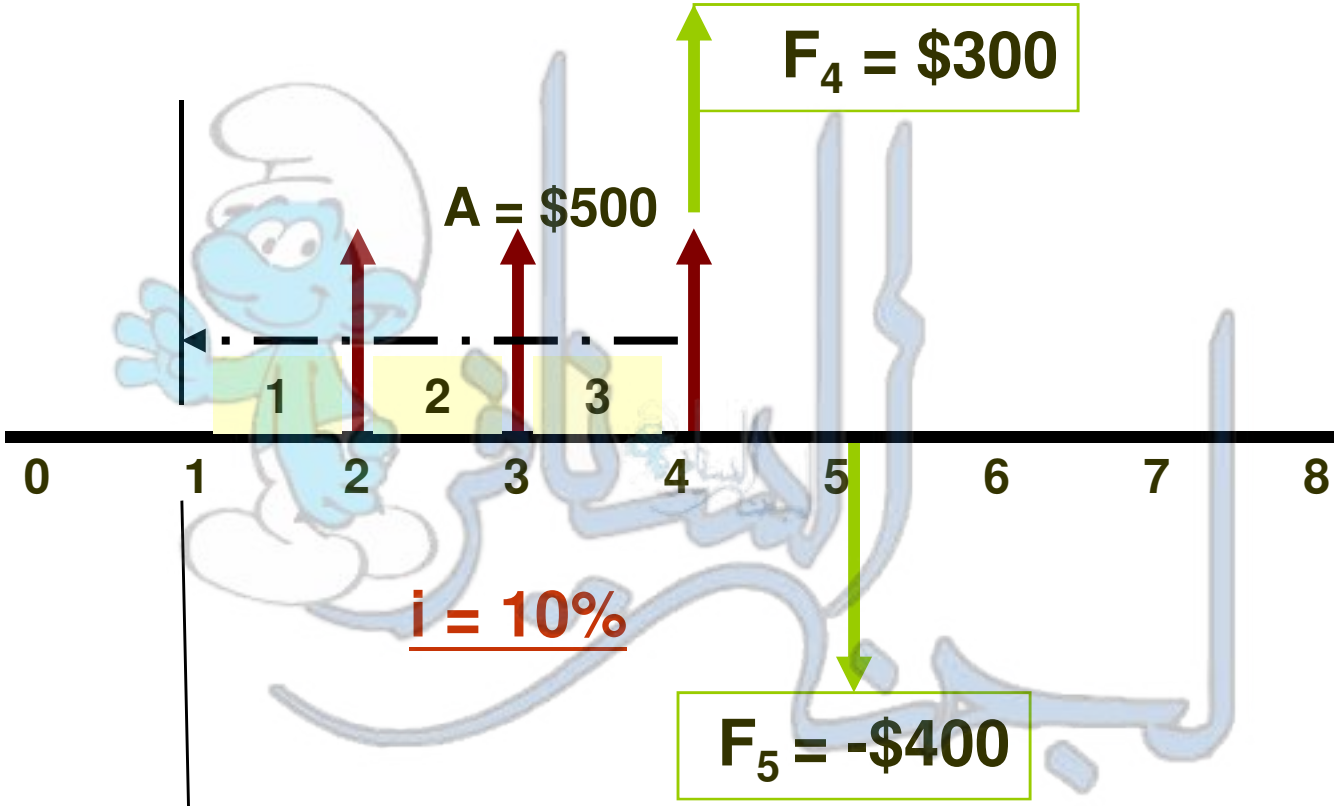
Series with Other cash flows

- Consider:



•Find the PW at t = 0 and FW at t = 8 for this cash flow

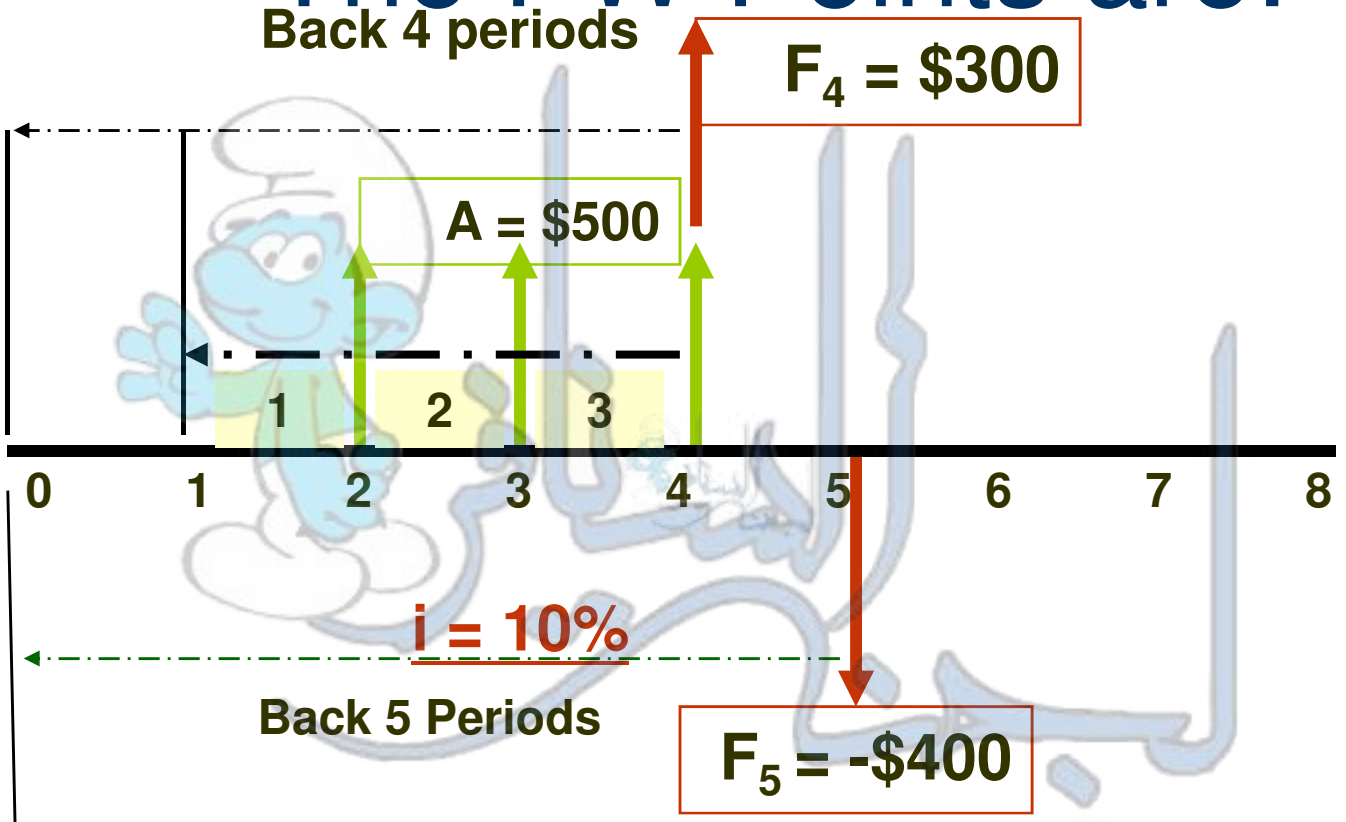
The PW Points are:



t = 1 is the PW point for the $500 annuity; "n" = 3

تم التحميل من شبكة المهندس المسلم
www.muslimengineer.info

The PW Points are:



t = 0 is the PW point for the two other single cash flows

Write the Equivalence Statement

P = $500(P/A,10%,3)(P/F,10%,1)

+

$300(P/F,10%,4)

-

400(P/F,10%,5)

Substituting the factor values into the equivalence expression and solving….

تم التحميل من شبكة المهندس المسلم
www.muslimengineer.info

Substitute the factors and solve

P = $500(2.4869)(0.9091)

+

$300(0.6830)

-

400(0.6209)

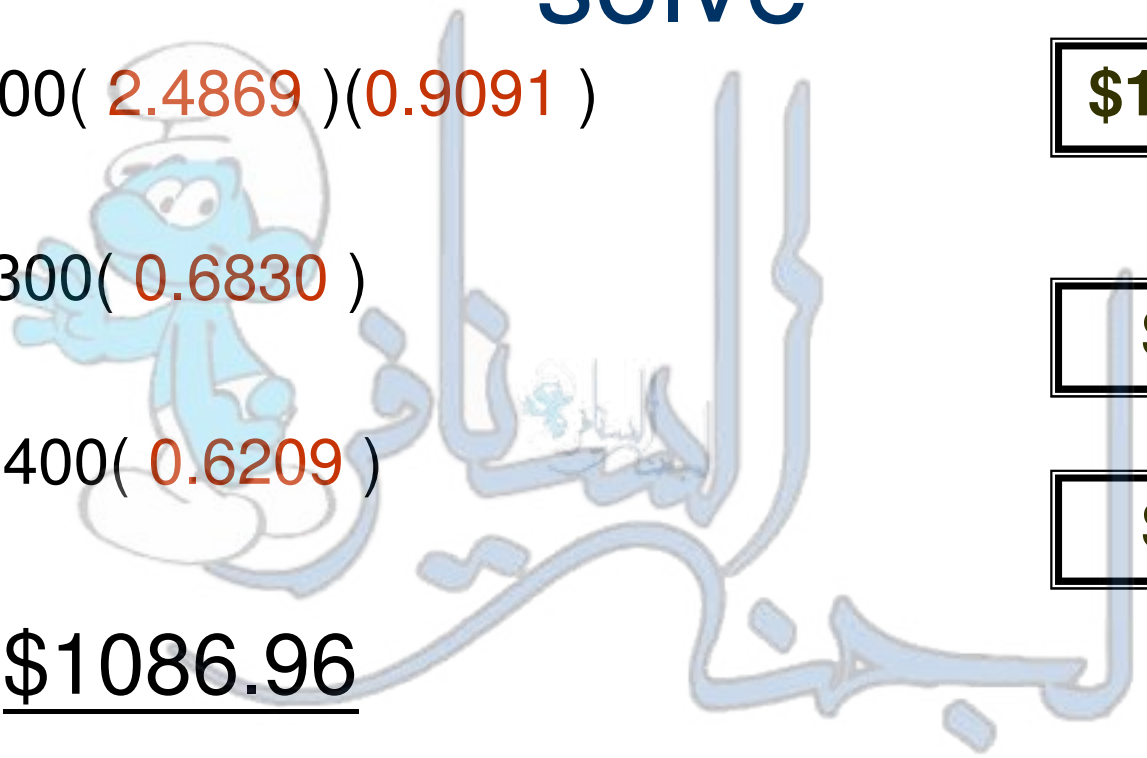
=

$1086.96

$1,130.42

$204.90

$248.36



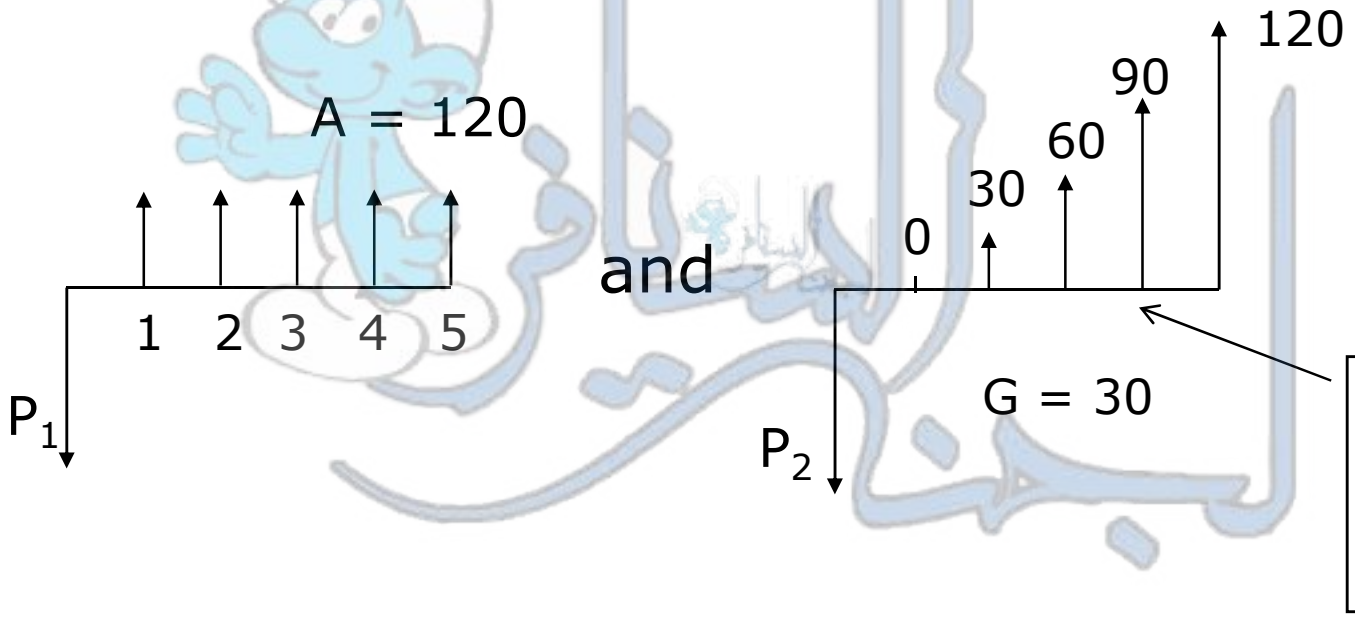
Arithmetic Gradient Series

- An arithmetic gradient is a cash flow series that either increases or decreases by a constant amount each period
- The **base amount** A_1 **(A)** is the uniform-series amount that begins in period 1 and ***extends*** through period n.
- Starting with the **second** period, each payment is greater (or smaller) than the previous one by a constant amount referred to as the **arithmetic gradient G**
- G can be positive or negative.

تم التحميل من شبكة المهندس المسلم
www.muslimengineer.info

Arithmetic Gradient

We break up the cash flows into two components:



Standard Form Diagram for Arithmetic Gradient: n periods and n-1 nonzero flows in increasing order

$P_1 = A\,(P/A,5\%,5) = 120\,(P/A,5\%,5) = 120\,(4.329) = 519$

$P_2 = G\,(P/G,5\%,5) = 30\,(P/G,5\%,5) = 30\,(8.237) = 247$

$P = P_1 + P_2 = \$766.$

Note: 5 and not 4. Using 4 is a *common mistake*.

Arithmetic Gradient

$$F = G(1+i)^{n-2} + 2G(1+i)^{n-3} + \ldots + (n-2)G(1+i)^{1} + (n-1)G(1+i)^{0}$$

$$F = G[(1+i)^{n-2} + 2(1+i)^{n-3} + \ldots + (n-2)(1+i)^{1} + n-1]$$

$$(1+i)F = G[(1+i)^{n-1} + 2(1+i)^{n-2} + 3(1+i)^{n-3} + \ldots + (n-1)(1+i)^{1}]$$

$$iF = G[(1+i)^{n-1} + (1+i)^{n-2} + (1+i)^{n-3} + \ldots + (1+i)^{1} - n + 1] =$$

$$= G[(1+i)^{n-1} + (1+i)^{n-2} + (1+i)^{n-3} + \ldots + (1+i)^{1} + 1] - nG =$$

$$= G(F/A, i, n) - nG = G[(1+i)^{n}-1]/i - nG$$

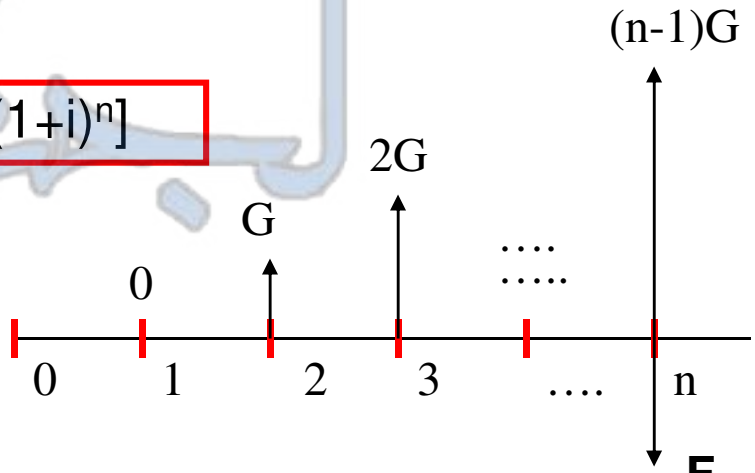
$$F = G[(1+i)^{n}-in-1]/i^{2}$$

$$P = F(P/F, i, n) = G[(1+i)^{n}-in-1]/[i^{2}(1+i)^{n}]$$

$$A = F(A/F, i, n) =$$

$$= G[(1+i)^{n}-in-1]/i^{2} \times i/[(1+i)^{n}-1]$$

$$A = G[(1+i)^{n}-in-1]/[i(1+i)^{n}-i]$$



تم التحميل من شبكة المهندس المسلم
www.muslimengineer.info

Arithmetic Gradient

Arithmetic Gradient Uniform Series

$(P/G,i,n) = \{ [(1+i)^n - i\,n - 1] / [i^2 (1+i)^n] \}$ $=1/(G/P,i,n)$

Arithmetic Gradient Present Worth

$(A/G,i,n) = \{ (1/i) - n/ [(1+i)^n - 1] \}$ $=1/(G/A,i,n)$

$(F/G,i,n) = G\,[(1+i)n\text{-}in\text{-}1]/i2$ $=1/(G/F,i,n)$

$(P/G,5\%,5) =$

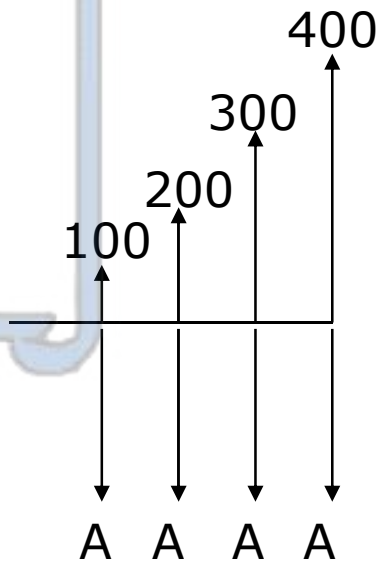
$= \{[(1+i)^n - i\,n - 1]/[i^2 (1+i)^n]\}$

$= \{[(1.05)^5 - 0.25 - 1]/[0.05^2 (1.05)^5]\}$

$= 0.026281562/0.003190703 = 8.23691676.$

Arithmetic Gradient

Example 4-6. Maintenance costs of a machine start at $100 and go up by **$100** each year for **4 years**. What is the equivalent uniform annual maintenance cost for the machinery if **i = 6%**.

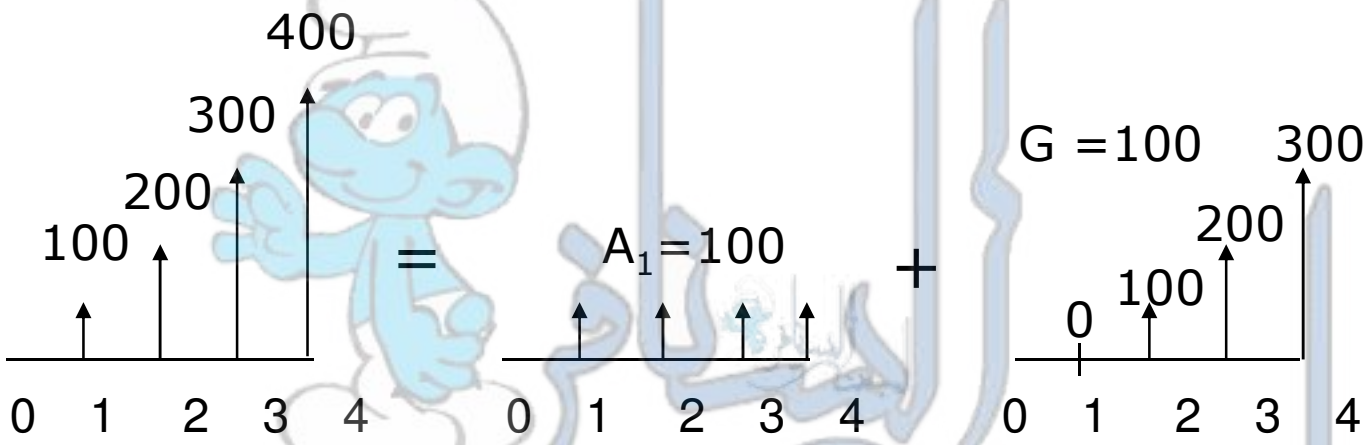


This is not in the standard form for using the gradient equation, because the year-one cash flow is not zero.

We reformulate the problem as follows.

تم التحميل من شبكة المهندس المسلم
www.muslimengineer.info

Arithmetic Gradient



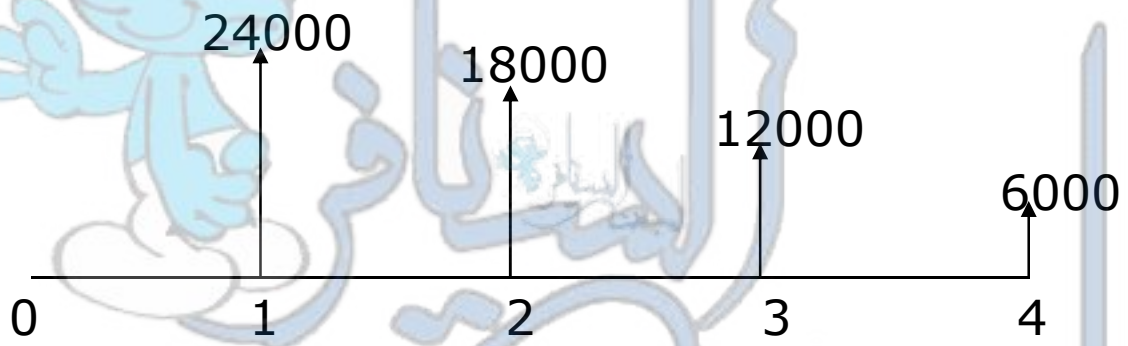
The second diagram is in the form of a $100 uniform series. The last diagram is now in the standard form for the gradient equation with n = 4, G = 100.

$$A = A_1 + G\,(A/G,6\%,4) = 100 + 100\,(1.427) = \$242.70$$

Arithmetic Gradient

Example

With i = **10%,** n = **4**, find an equivalent uniform payment A' for

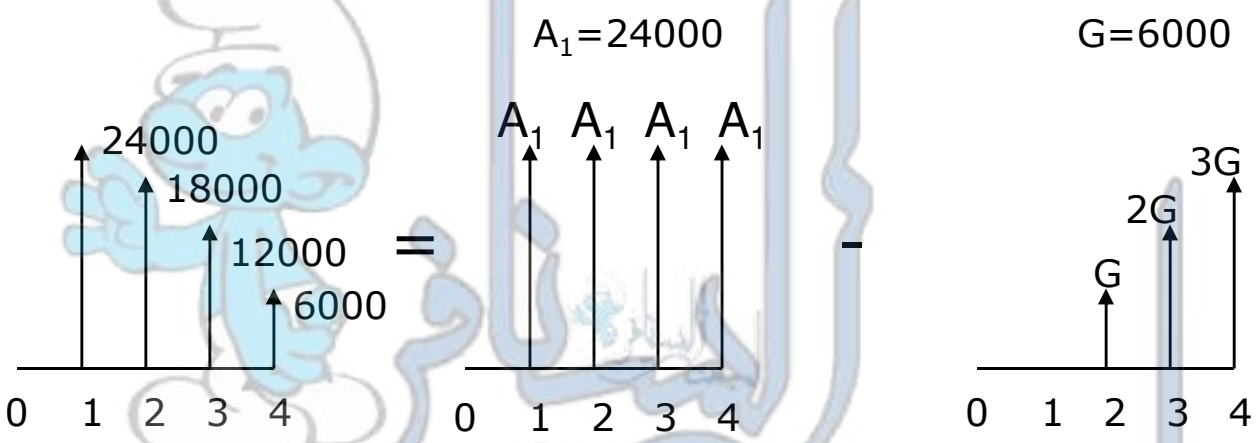


This is a problem with decreasing costs instead of increasing costs.

The cash flow can be rewritten as the DIFFERENCE of the following two diagrams, the second of which is in the standard form we need, the first of which is a series of uniform payments.

تم التحميل من شبكة المهندس المسلم
www.muslimengineer.info

Arithmetic Gradient



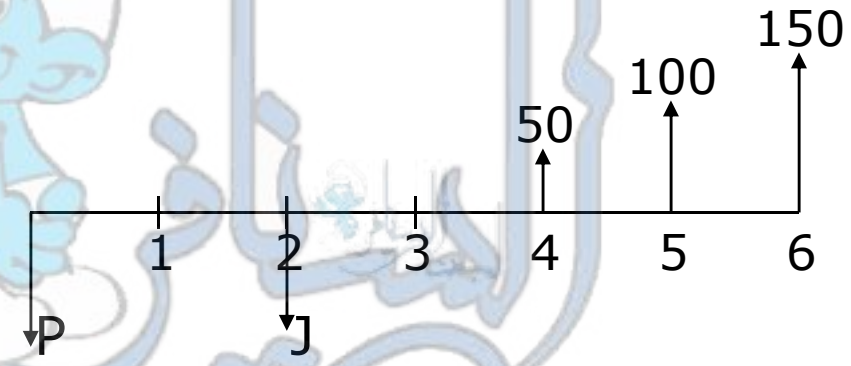
$$A = A_1 - G(A/G,10\%,4) =$$

$$= 24000 - 6000\,(A/G,10\%,4) =$$

$$= 24000 - 6000(1.381) = 15{,}714.$$

Arithmetic Gradient

Example Find P for the following diagram with i = 10%.



This is not in the standard form for the arithmetic gradient. However, if we insert a "present value" J at the end of year 2,
the diagram ***from that point on*** IS in standard form.

Thus:

J = 50 (P/G,10%,4) = 50 (4.378) = 218.90

P = J (P/F,10%,2) = 218.90 (0.8264) = $180.9

تم التحميل من شبكة المهندس المسلم
www.muslimengineer.info